



Hands-On: Signal Chain Noise

PDF of this deck:

<https://tinyurl.com/2026-adi-psc>

analog.com

Material List



- ADALM2000 unit with analog fly wires
 - upgraded to firmware 0.33
- ADALM-MMSC board
 - MAX32666FTHR controller board
- Attendee Laptop Requirements:
 - Windows 10, Windows 11
 - Mac OSX 14.8, or 15.7
- Two spare USB-A ports, or USB hub
- INSTALL Scopy 2.2.0:
<https://github.com/analogdevicesinc/scopy/releases/tag/v2.2.0>
Use default locations, DO install ADALM2000 drivers when prompted (may prompt for elevation/admin – accept)

Scopy Installation Guide:

https://developer.analog.com/docs/scopy/user_guide/index.html#installation

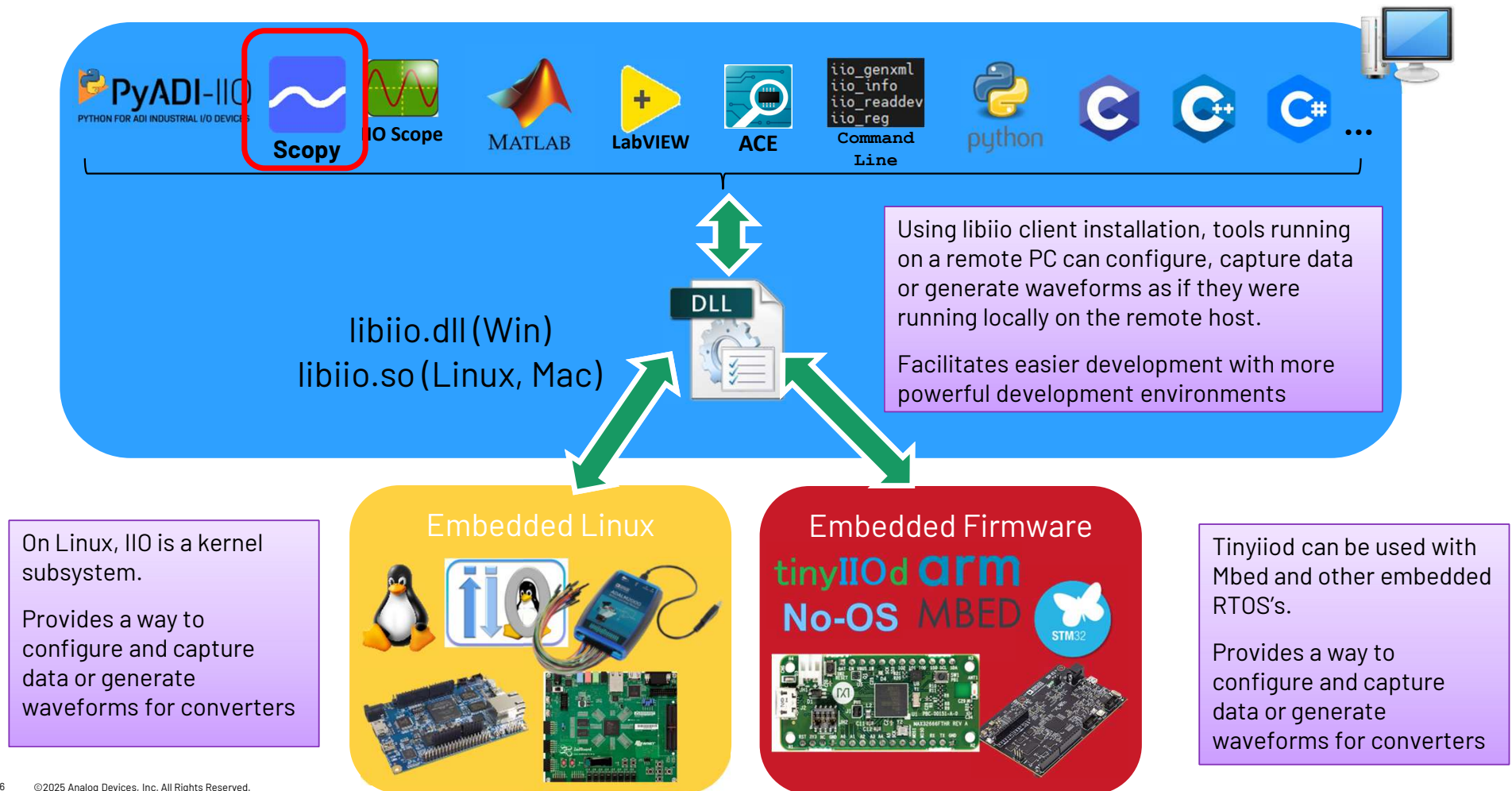
Objectives

- Measuring differential signals with ADALM2000 (M2k) Oscilloscope and ADC Plugin
- Demonstrating the capabilities of analog antialiasing and digital filter
- See tradeoffs of performance vs latency when implementing low pass filters

All of this while:

- Moving beyond complicated, expensive, drifts, 10th-order analog antialias filters
- Moving beyond power-hungry ADC driver circuits
- Moving beyond expensive FPGAs for data capture

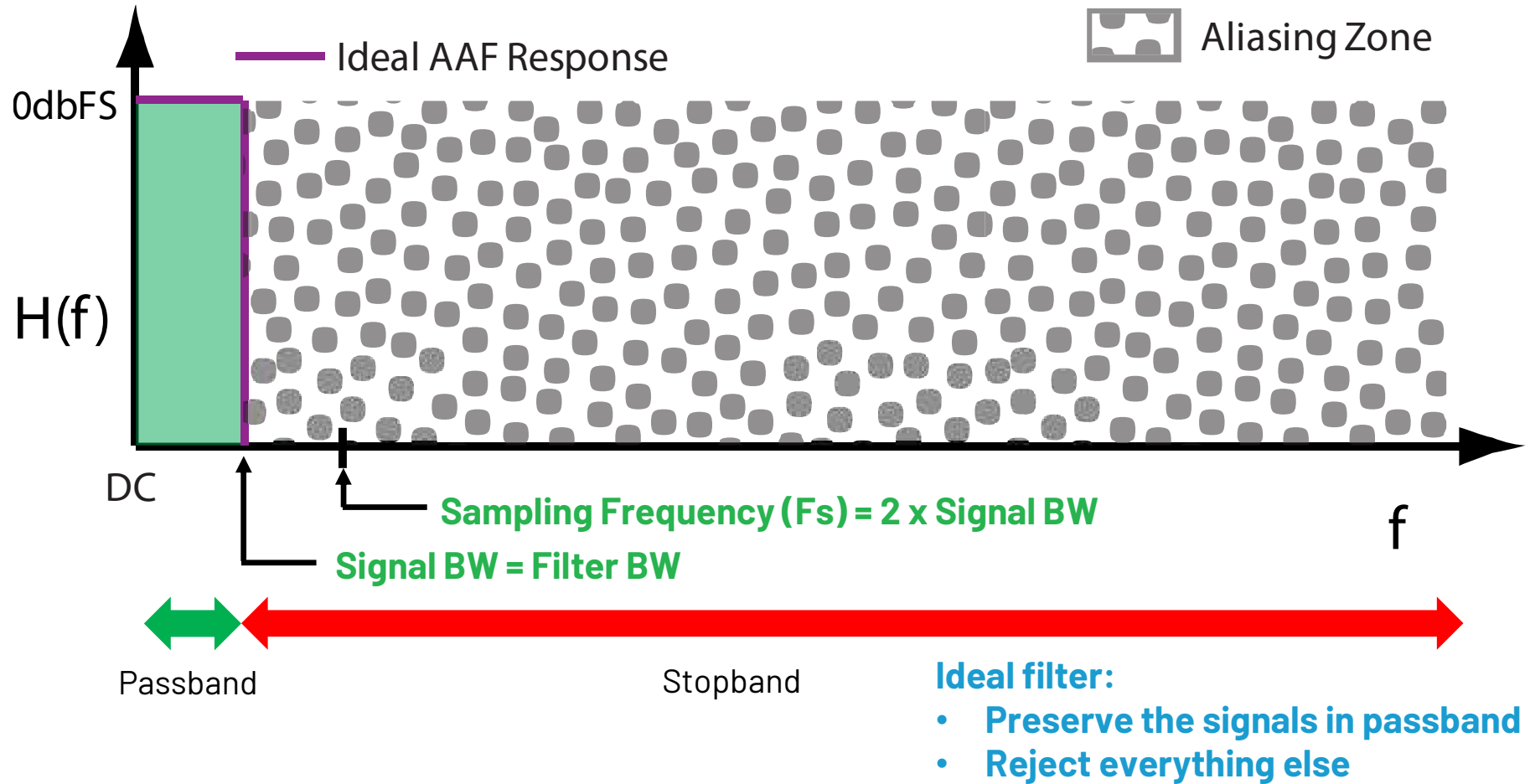
Flexible, IIO-based System Architecture



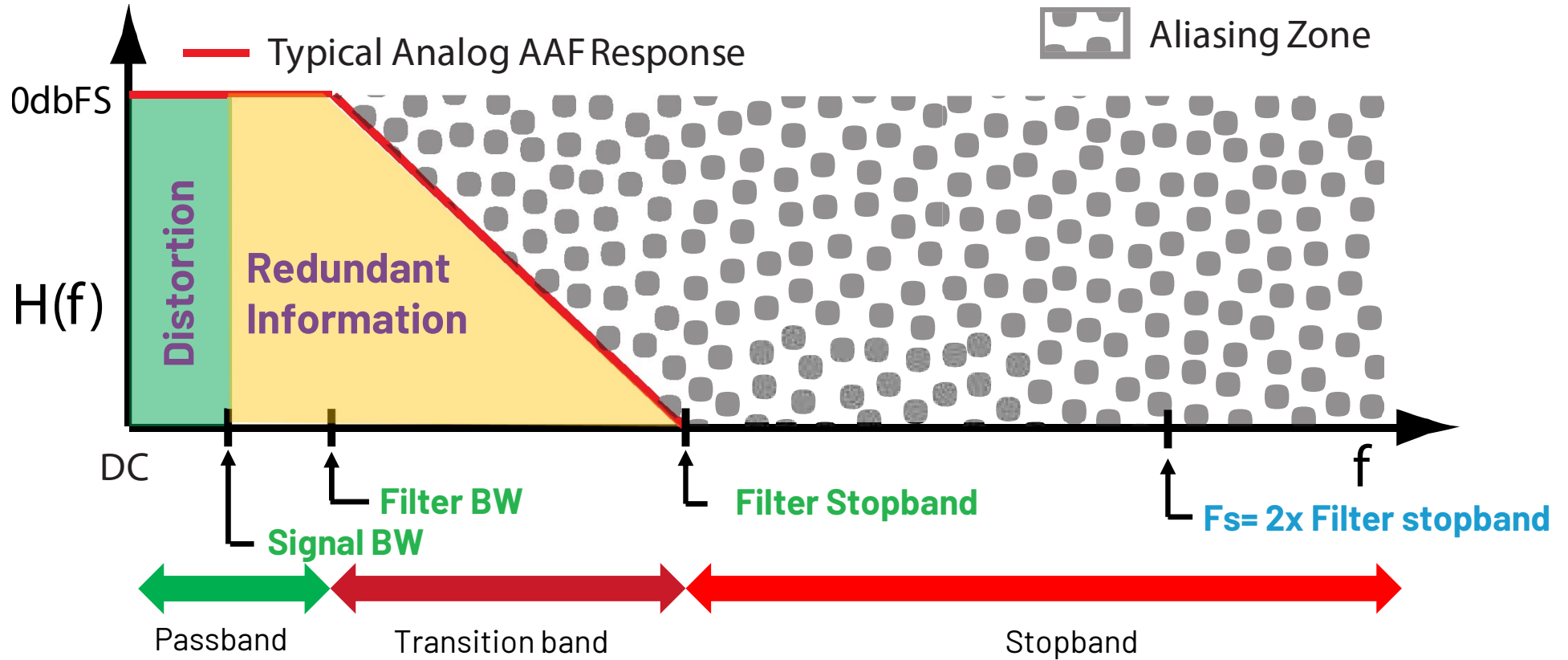


Analog Filter Design Challenges

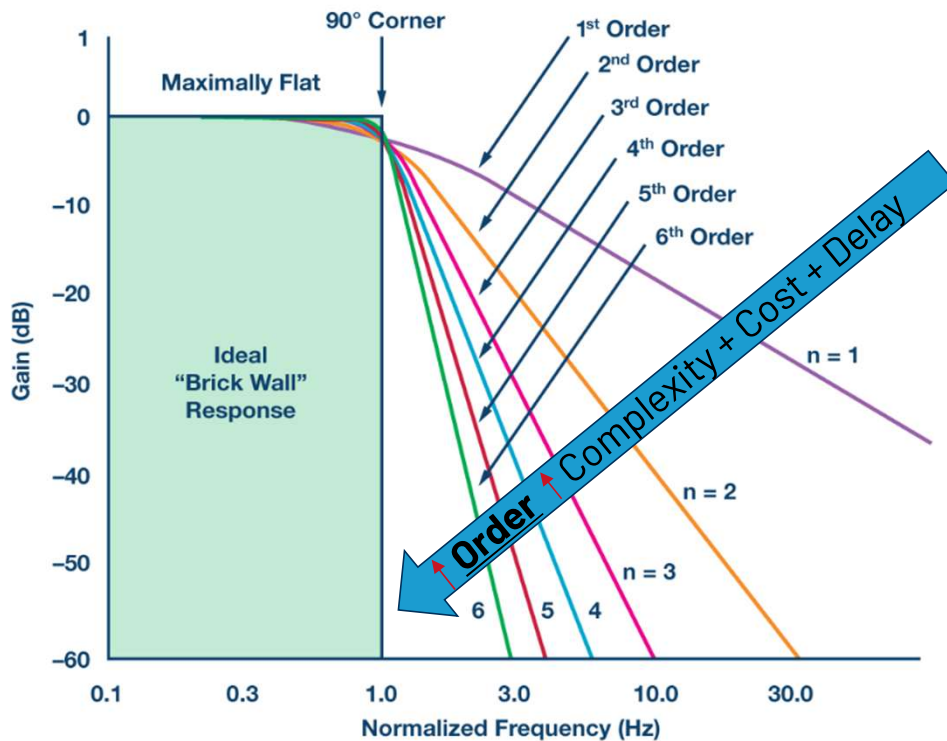
Frequency Response of Ideal Anti-Aliasing Filter (AAF)



Actual Anti-Aliasing Filter



Filter Order vs. Complexity vs. Noise vs. Cost



Filter Order

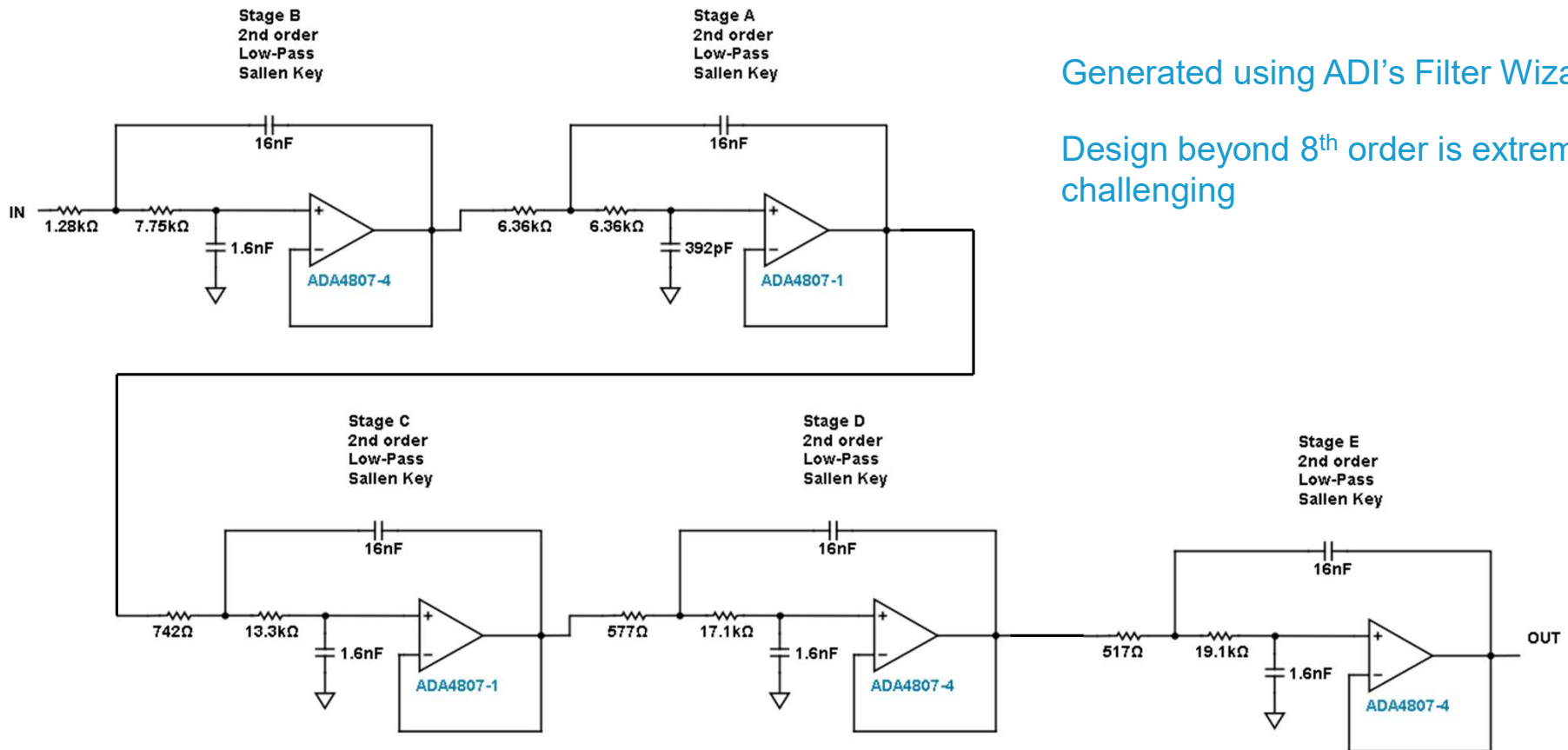
Higher the order:

- Noise ✓
 - Higher Alias Rejection = Better Noise & SNR
- Complexity & Cost ✗
 - Higher Design Complexity = Bigger Cost & Area & More Resources and Development Time
- Delay ✗
 - Higher Delay = Higher Latency in Real Time System

To Learn More About Analog Filter Design Go to:

- <https://www.analog.com/media/en/training-seminars/design-handbooks/basic-linear-design/chapter8.pdf>
- <https://www.analog.com/en/resources/analog-dialogue/articles/practical-filter-design-precision-adcs.html>

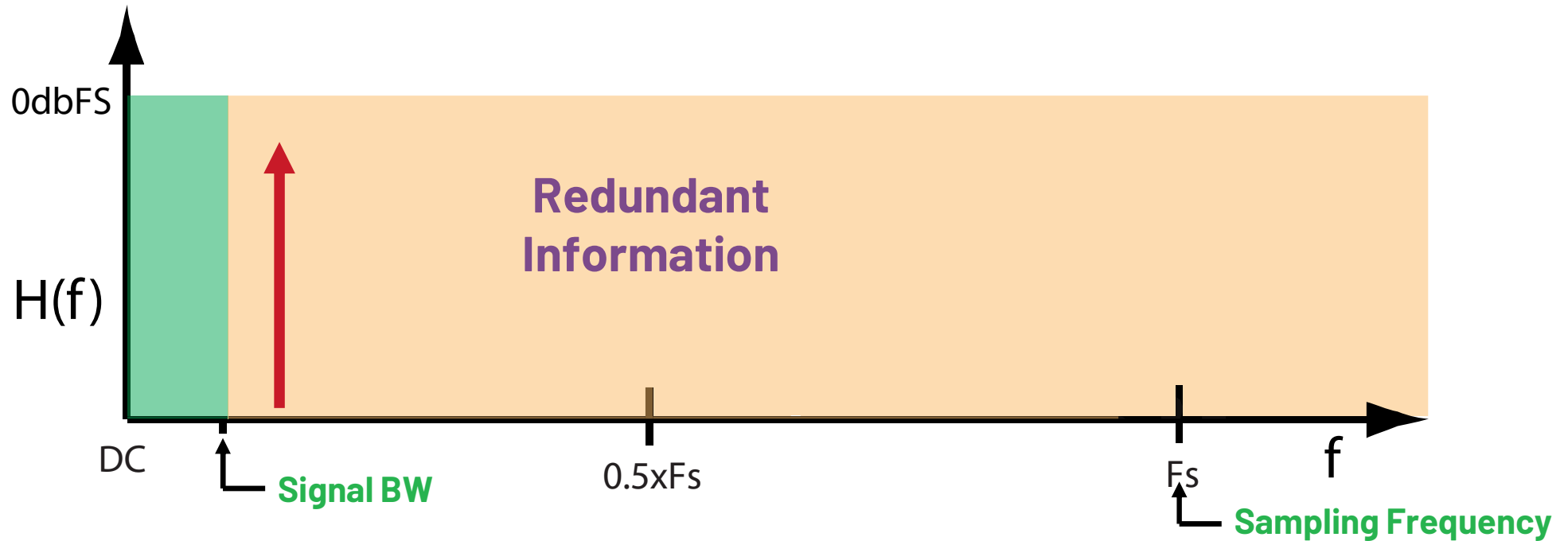
An Example of 10th Order Analog Low Pass Filter



Generated using ADI's Filter Wizard Tool

Design beyond 8th order is extremely challenging

Oversampling: Ratio of F_s to 2x signal BW



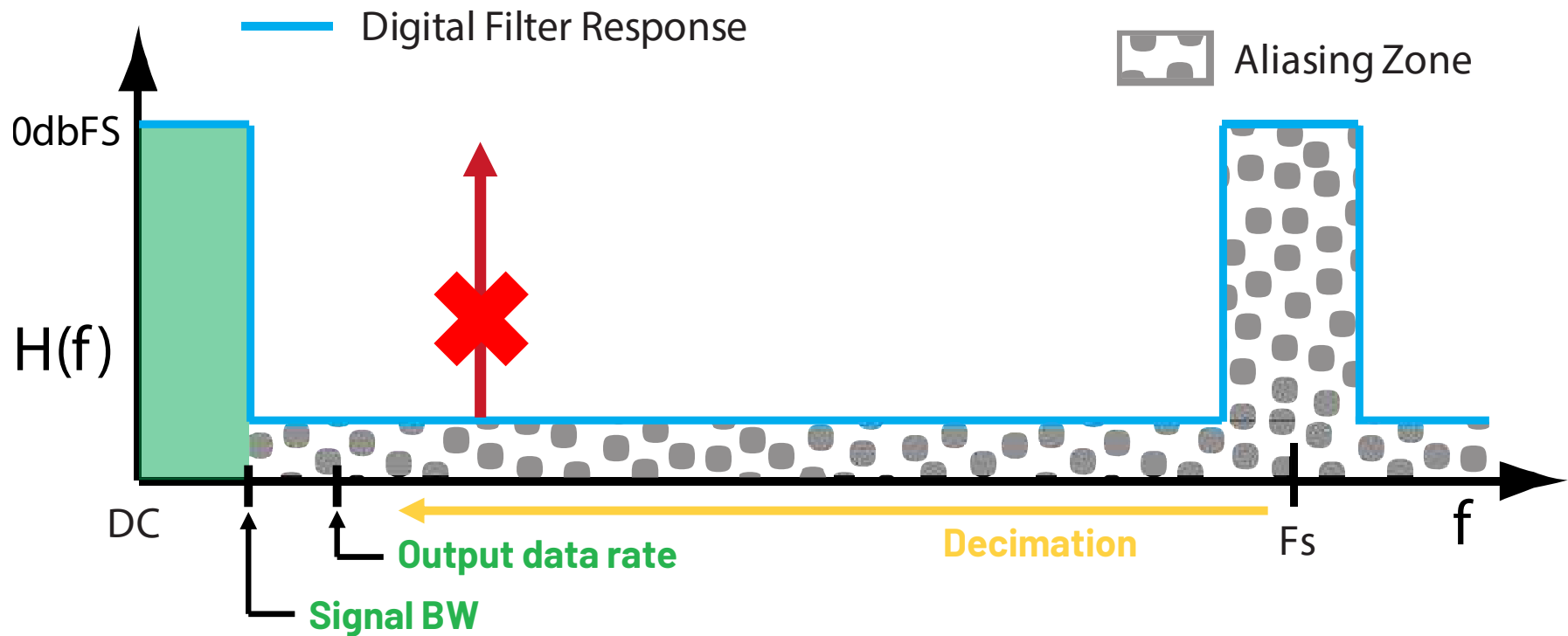
Pro:

- Differentiation of out of band signals
- ADC's own noise density is lower

Con:

- Redundant information in the output – higher data rate
- Only works in frequency analysis (integrated power (rms) is still the same)
- Frequency band around F_s is not protected.

Oversampling + Digital Filter



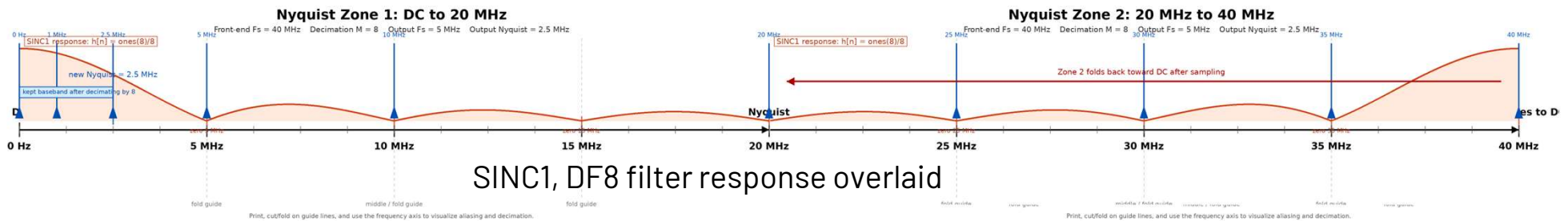
Digital filter + Decimation removes the redundant information from the output and keeps the output data rate low

Important Digital Filter Concepts

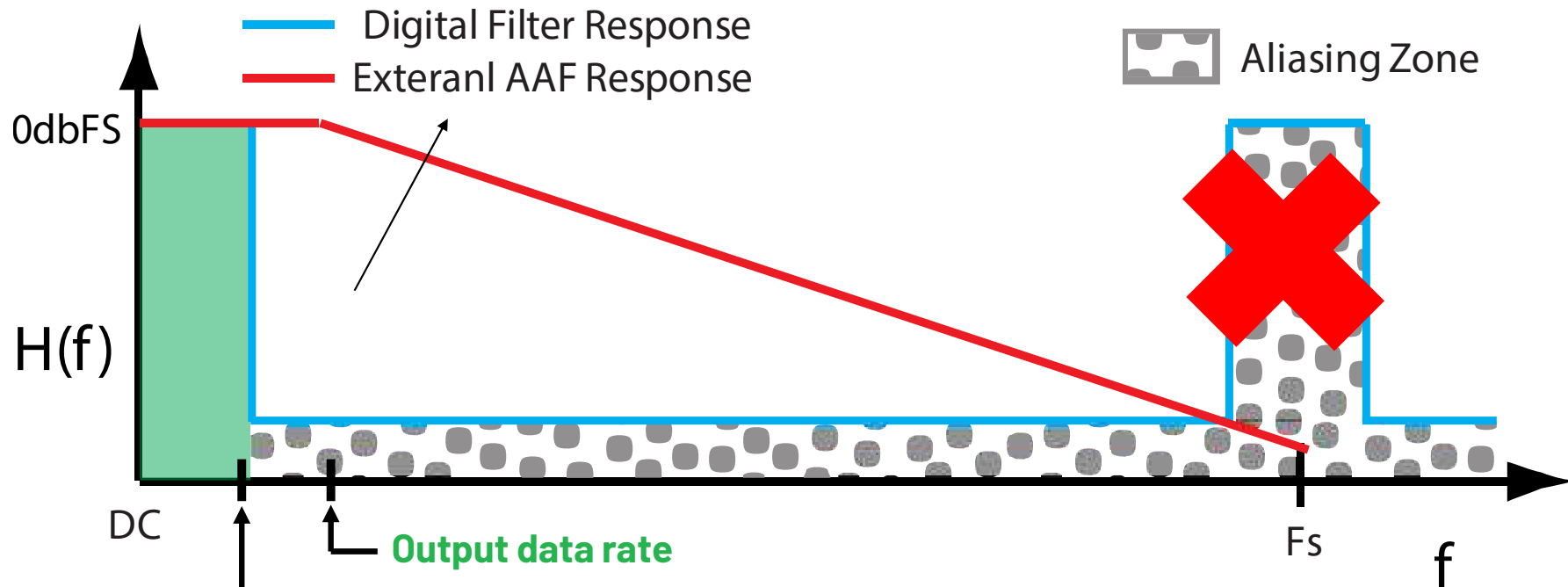


- Sampling faster “widens” Nyquist zones
 - Simpler analog filtering
 - But – you must process the extra data**
- Downsampling (reducing output data rate) “folds” Nyquist zones
 - Folded zones alias on top of one another

** Spoiler – we do this inside the AD4080!



Combining Analog and Digital Filters



Full out of band aliasing protection

Higher over sampling ratio => simpler analog filter design

Visit Wiki Page



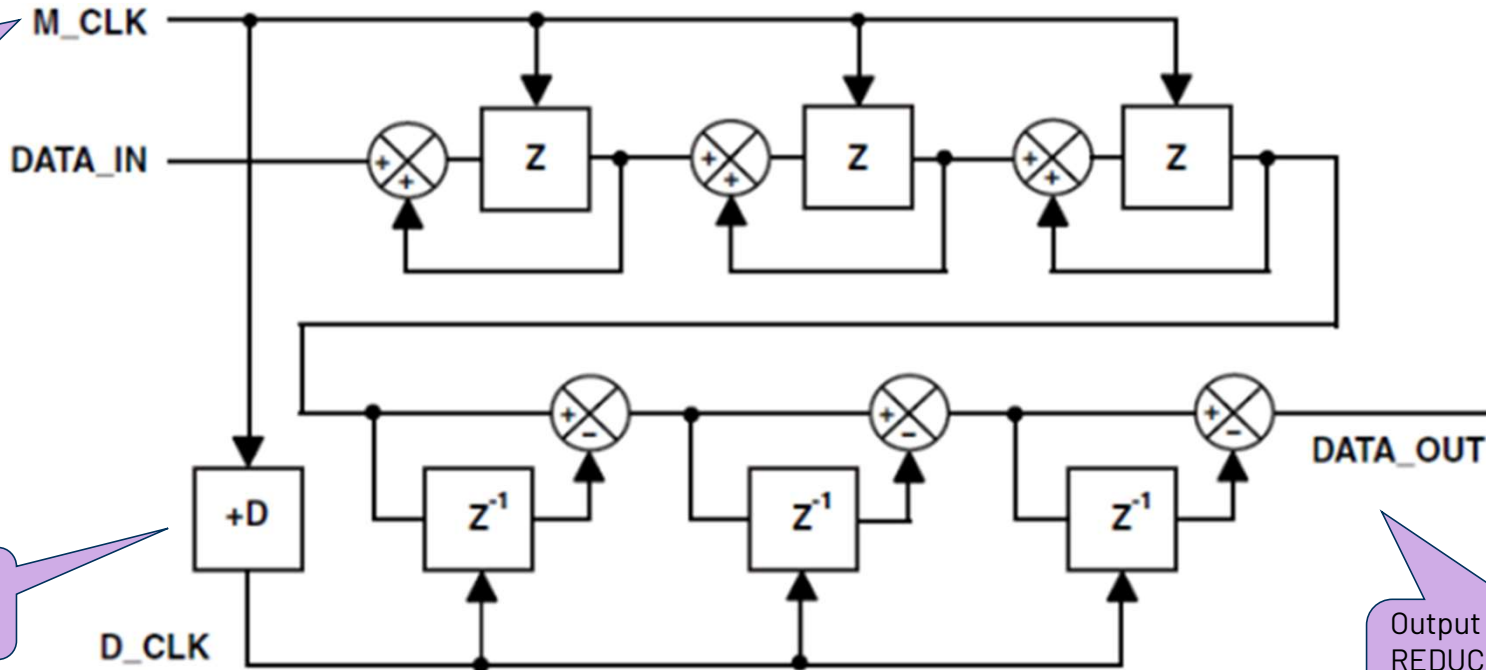
t.ly/ThG31

On-chip Digital Filter

Digital Filter Example Block Diagram (Sinc 3 CIC Filter)



AD4080:
40 Msps Max,
1.25 Msps Min



Downsample
Factor

Output Data Rate is
REDUCED (decimated)
by the Downsample
Factor

See:

IEEE TRANSACTIONS ON ACOUSTICS, SPEECH, AND SIGNAL PROCESSING, VOL. ASSP-29, NO. 2, APRIL 1981

155

An Economical Class of Digital Filters for Decimation
and Interpolation

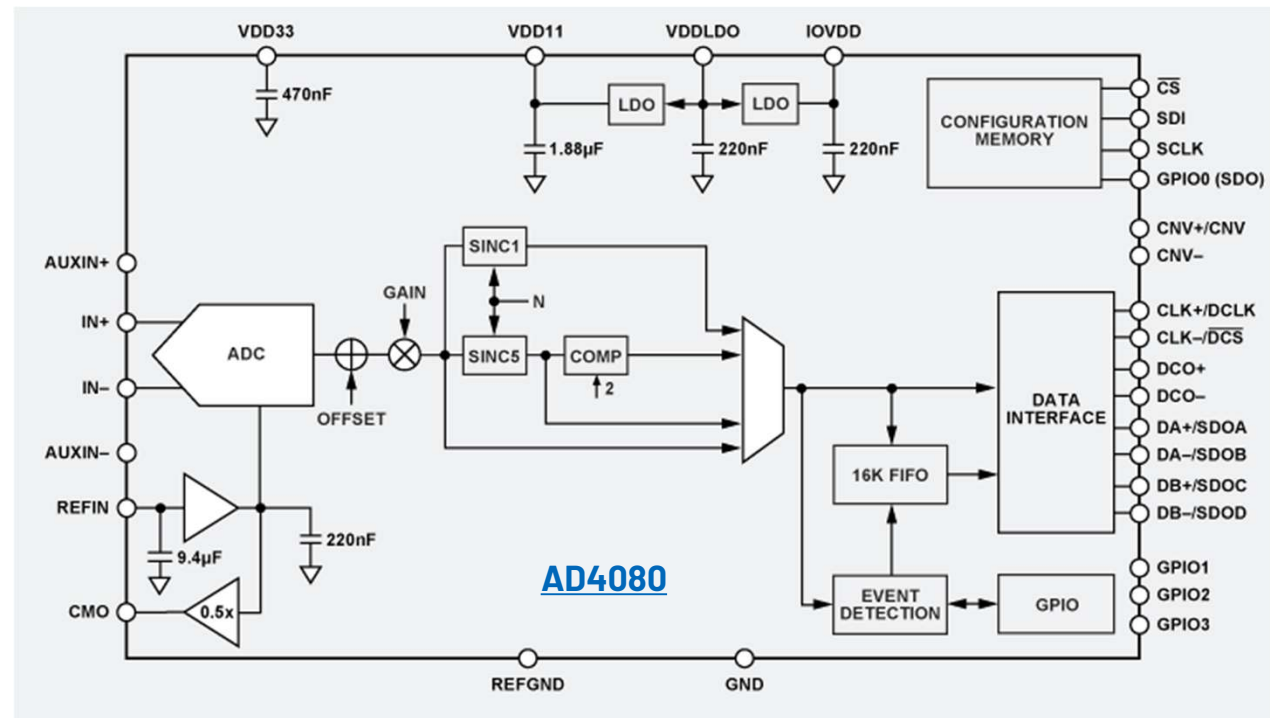
EUGENE B. HOGENAUER, MEMBER, IEEE

AD4080 40MSPS 20bit Differential SAR ADC



Key Benefits

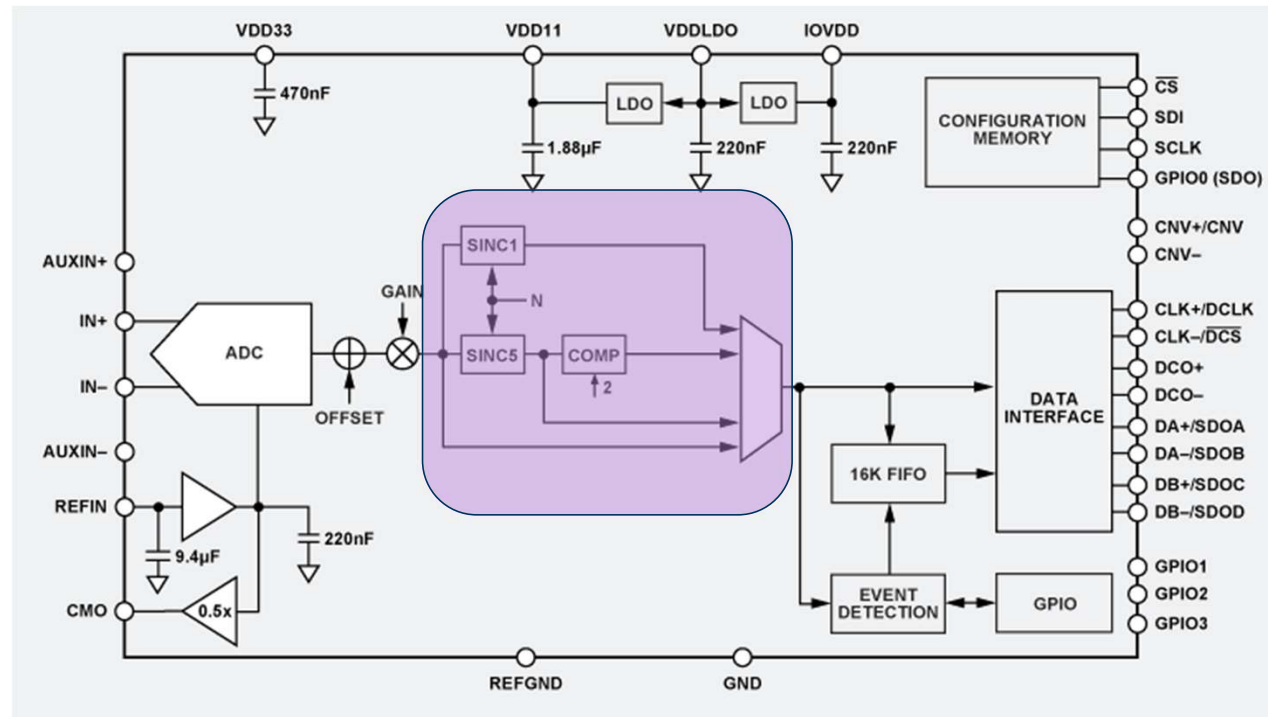
- Performance: Highest precision, lowest noise ADC at this speed for only **80mW**. Low latency **46.25 ns**.
- **Compact** and simplifies designs: Critical capacitors contained within a 5 mm × 5 mm BGA package.
- Flexibility: Programmable offset/gain correction, **digital filtering**, event trigger, FIFO, SPI/LVDS data.



AD4080 40MSPS 20bit Differential SAR ADC



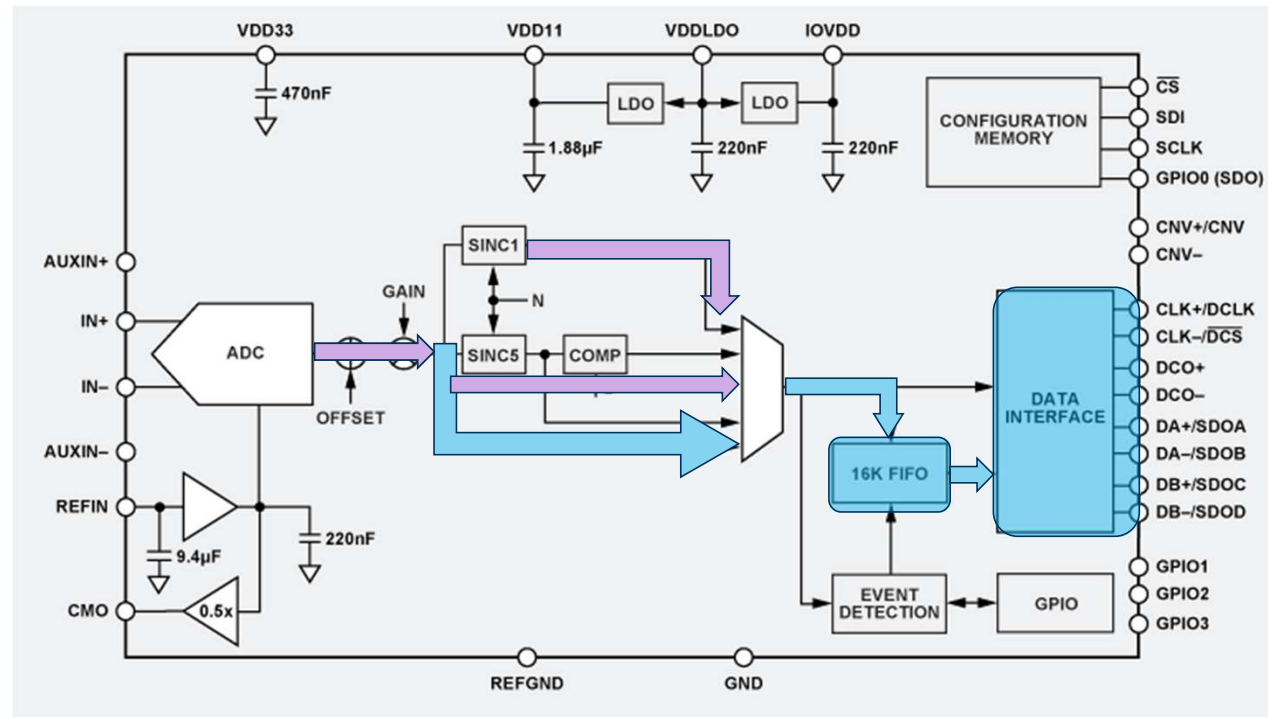
- The AD4080 includes the option of enabling an **integrated** digital filter for applications where **noise rejection** by bandwidth limiting is desired.
- These filters allow the user to **programmatically** control the noise bandwidth of their signal chain and also can offer benefits by reducing the amount filtering required in the analog front end.



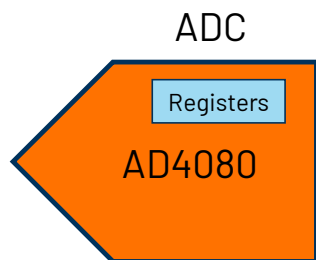
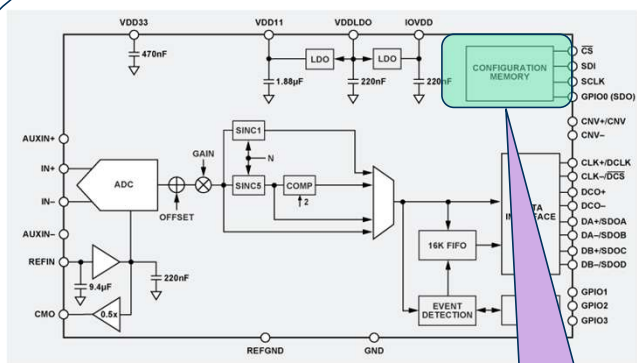
AD4080 40MSPS 20bit Differential SAR ADC



- There are two acquisition modes available:
- ✓ continuous fast data acquisition
- ✓ **noncontinuous bursts of data acquisition**
- In Burst Acquisition Mode allows data from ADC is automatically loaded to **FIFO**
- The data FIFO allows for a record of up to **16,384** data results to be captured per acquisition burst
- **SPI** can be used to read the results of the FIFO with any microcontroller (no need for costly FPGA)



Configure ADC Registers by setting IIO ADC Attributes

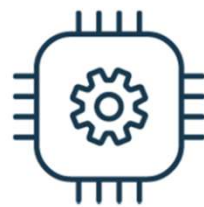


Direct read/writes to Registers

W: SPI Data

R: SPI Data

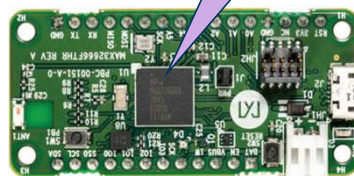
Read/Write to ADC Registers



Firmware

Firmware Already Programmed

uC

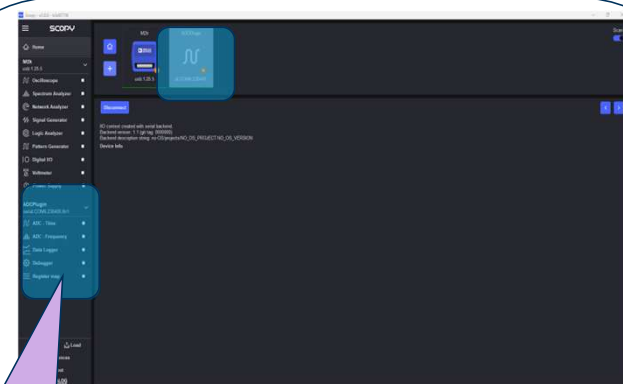


H
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FIRMWARE



S
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Configure Attributes via GUI Interface

W: USB Data

R: USB Data

PC



Configure ADC IIO Attributes

AD4080 Digital Filters Frequency Responses



Sinc1

Wider BW, not optimized for passband flatness

Sinc Filter = Low Latency

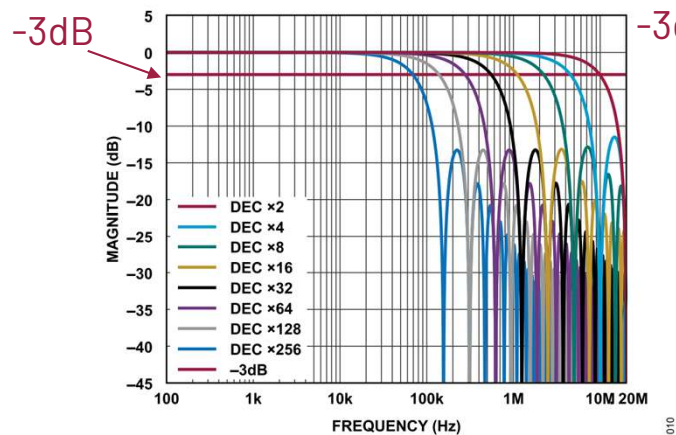


Figure 9. Sinc1 Filter Response, $f_s = 40$ MHz (DEC x Means Decimate By)

Sinc5

Improved passband flatness with reduced BW

Sinc Filter = Low Latency

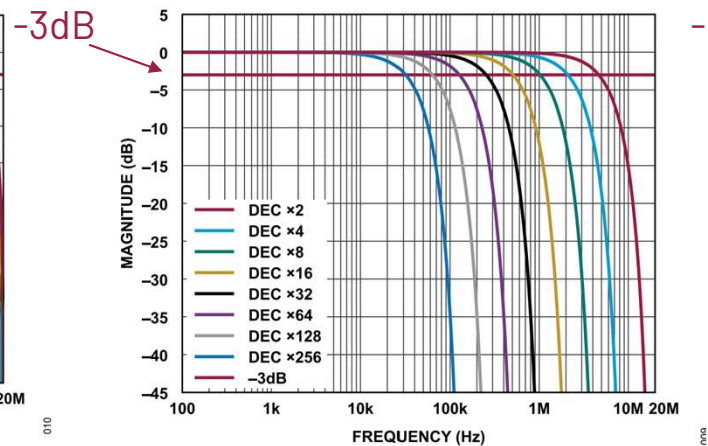


Figure 10. Sinc5 Filter Response, $f_s = 40$ MHz

Brickwall (Sinc5 + Comp)

Narrow Transition Band, +/- 0.1dB ripple in passband

Flat Passband = High Latency

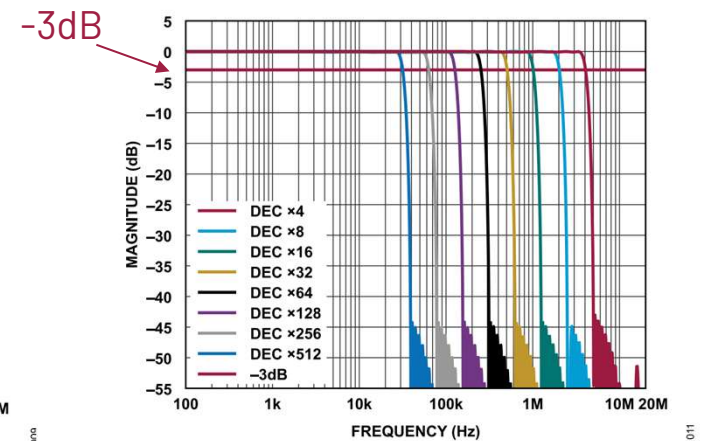


Figure 11. Sinc5 + Compensation Filter Response, $f_s = 40$ MHz

Higher Order = Sharper Roll - Off

Digital Filters -3dB BW Setup – Flexible Adjustment



Filter Type	SINC_DEC_RATE	Decimation	-3 dB Bandwidth
Sinc5	0000	2	$0.117 \times f_S$
Sinc5	0001	4	$0.0525 \times f_S$
Sinc5	0010	8	$0.0256 \times f_S$
Sinc5	0011	16	$0.0127 \times f_S$
Sinc5	0100	32	$0.0064 \times f_S$
Sinc5	0101	64	$0.0032 \times f_S$
Sinc5	0110	128	$0.0016 \times f_S$
Sinc5	0111	256	$0.0008 \times f_S$

Filter Bandwidth can be easily adjusted by selecting equivalent decimation rate which sets the SINC_DEC_RATE register to the right value.

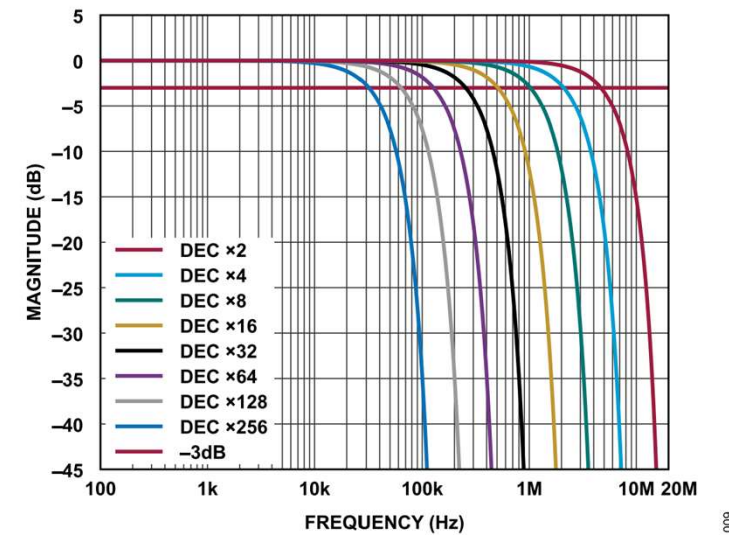
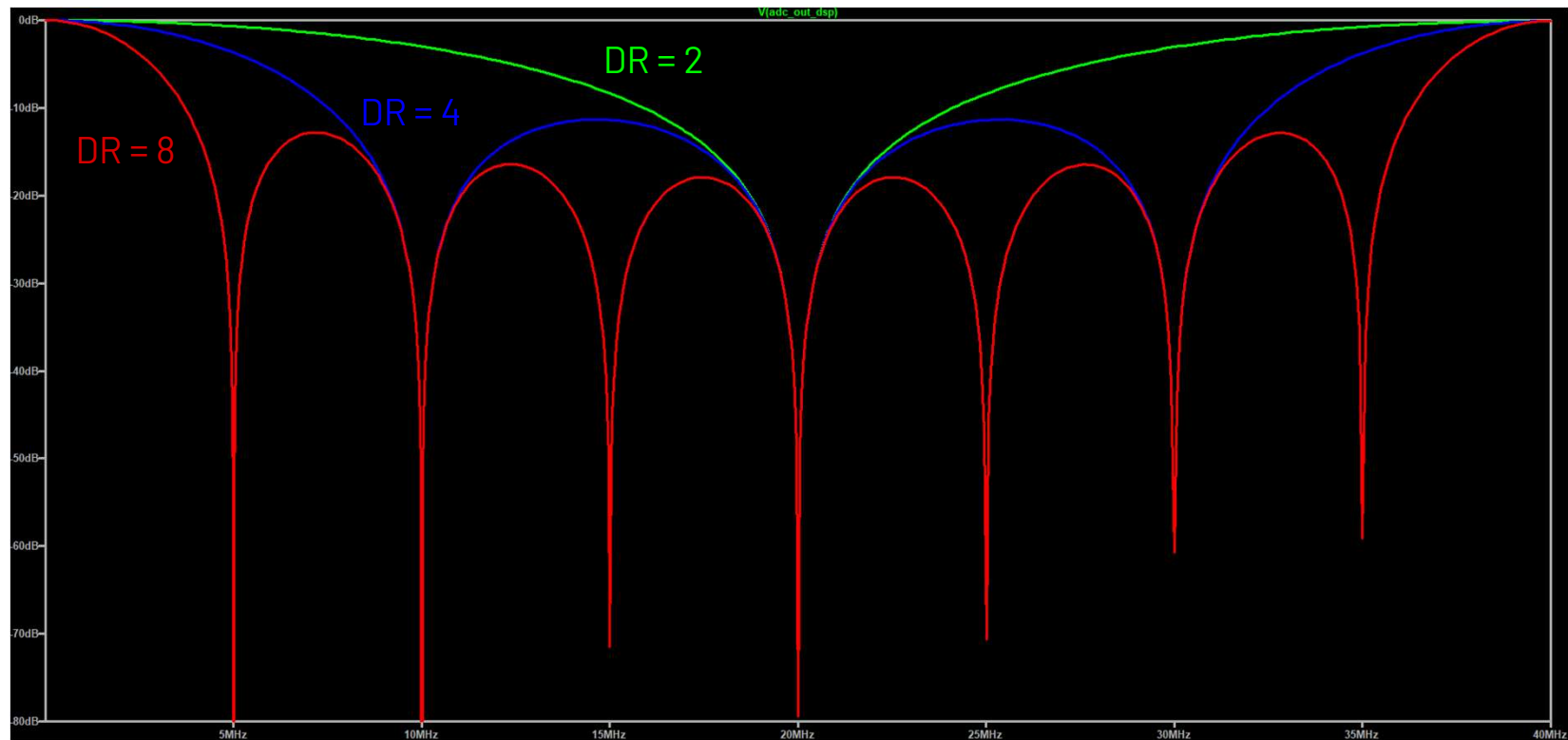


Figure 10. Sinc5 Filter Response, $f_S = 40 \text{ MHz}$

BW vs. Decimation Rate – Effect on Total Noise



Higher Decimation Rate (Oversampling Rate) = Better Noise Attenuation

Oversampling (DR) vs. Total Noise in BW



Decimation	-3 dB Bandwidth
2	$0.25 \times f_s$

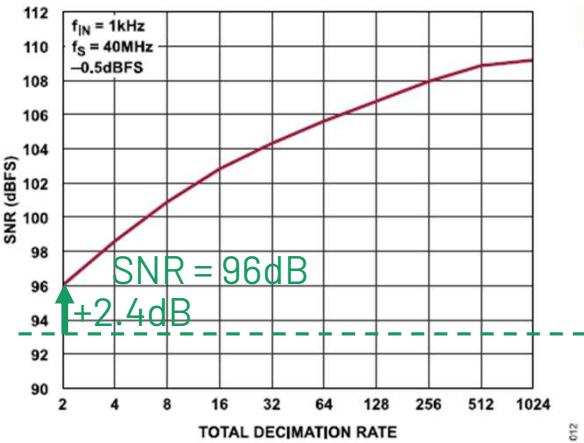
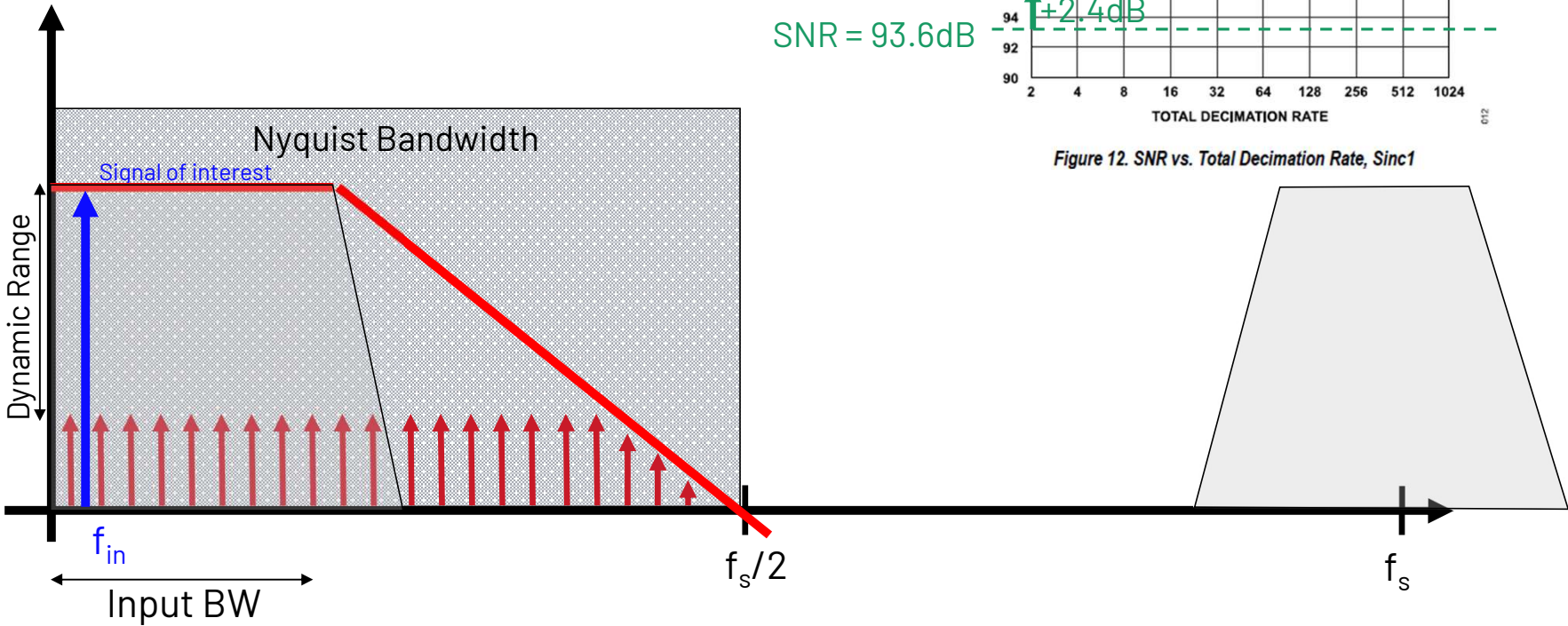
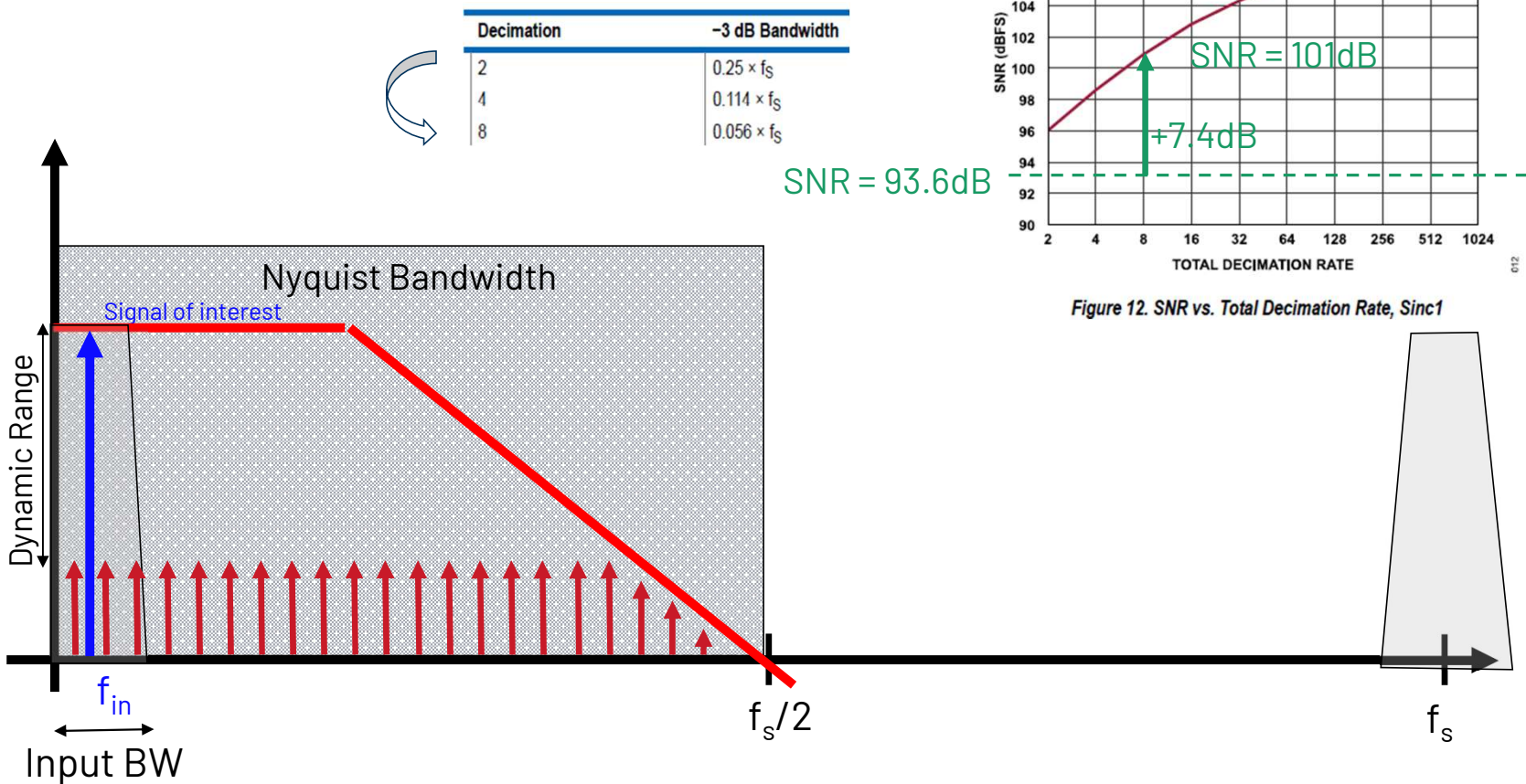


Figure 12. SNR vs. Total Decimation Rate, Sinc1

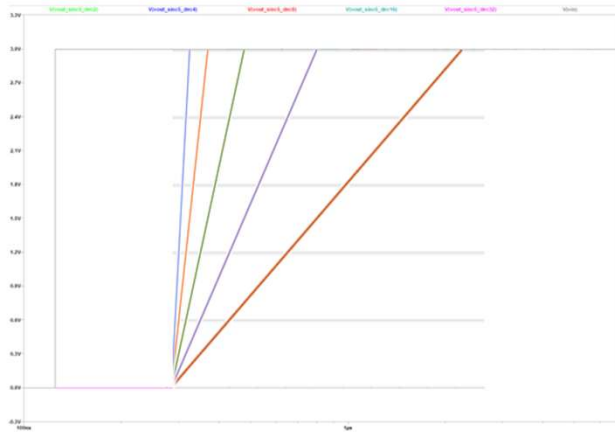
Oversampling (DR) vs. Total Noise in BW



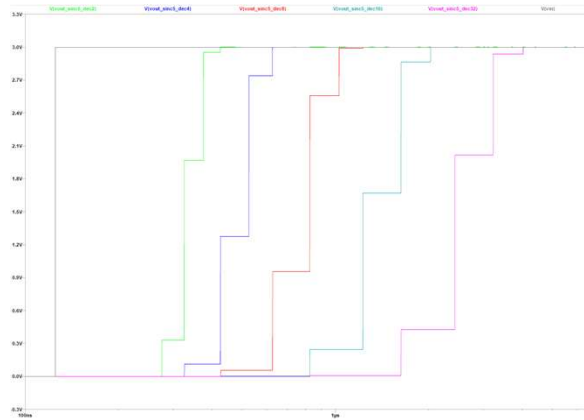
AD4080 Digital Filters Step Responses (Time Domain)



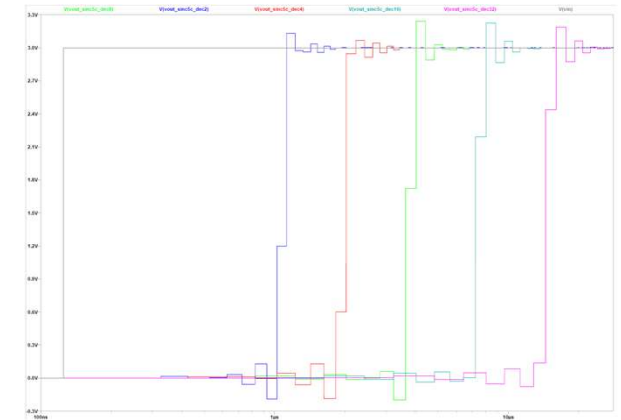
Sinc1



Sinc5



Sinc5 + Compensation



Higher Order = Higher Delay (Group Delay)



ADALM-MMSC Signal Chain Overview

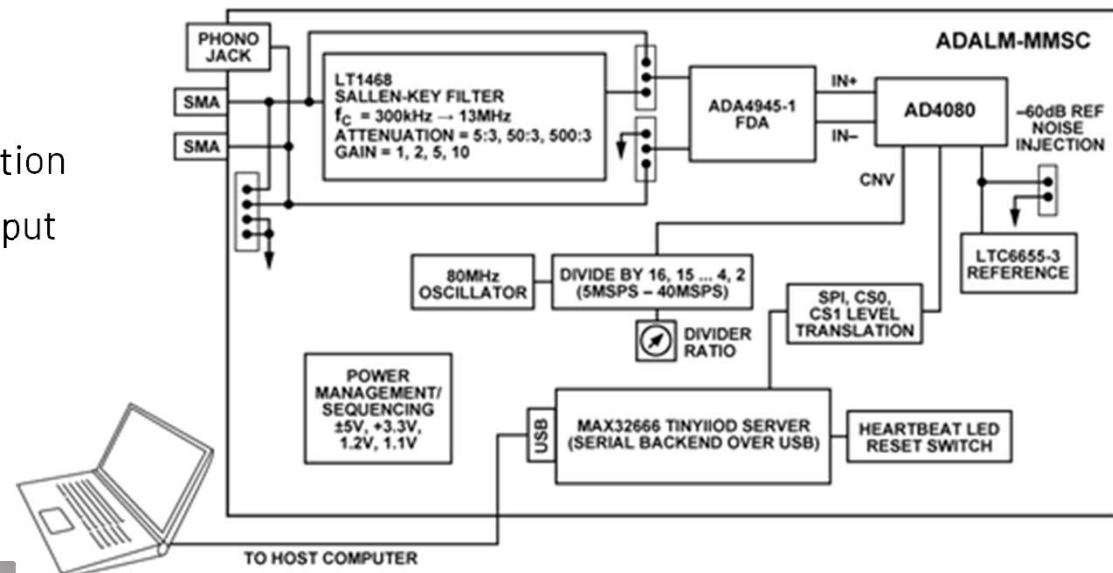
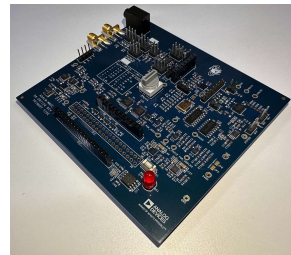
ADALM-MMSC LEARNING BOARD

"Analog Devices Active Learning Module – Mixed-Mode Signal Chain"

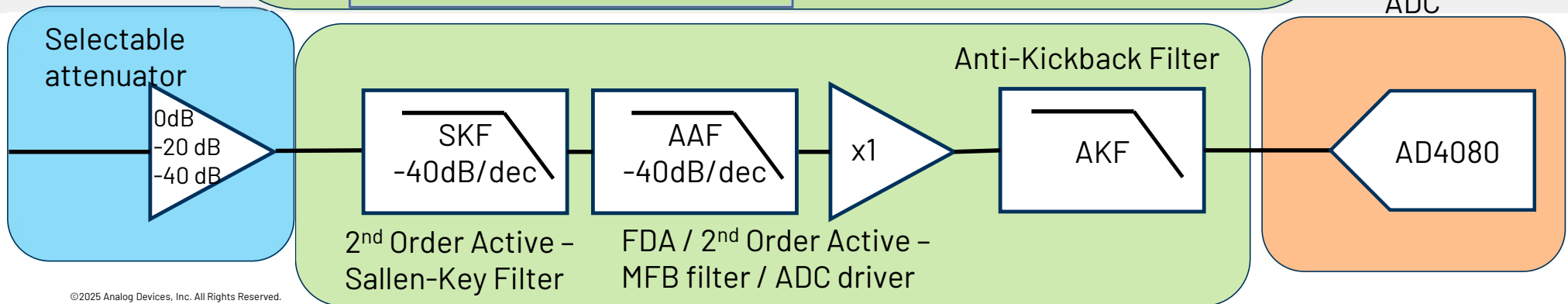
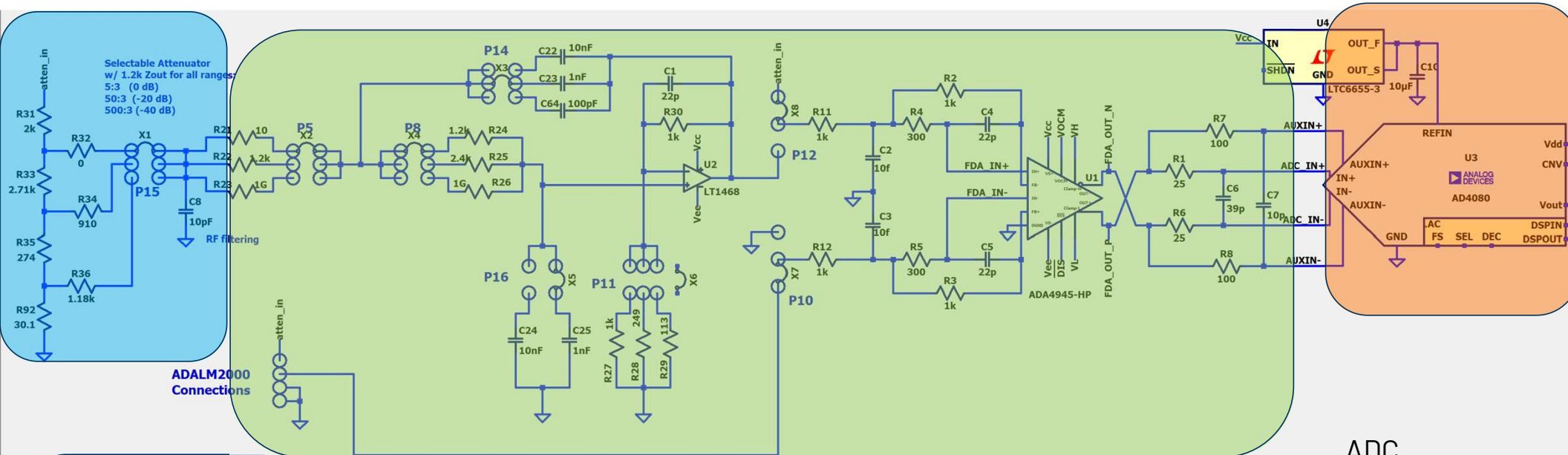
Key Benefits

- Full featured learning board for the [AD4080](#) with signal chain, power, and computer interface solution
- Single differential channel and common-mode input available through board SMA connectors
- Flexible analog front end (two stages) with configurable analog filter
- Adjustable clock rate from 5 to 40 MSPS

- Link to board: [ADALM-MMSC](#)



MMSC Signal Chain – Attenuator, Analog Filtering, Gain, FDA Buffer Stages



Visit Wiki Page



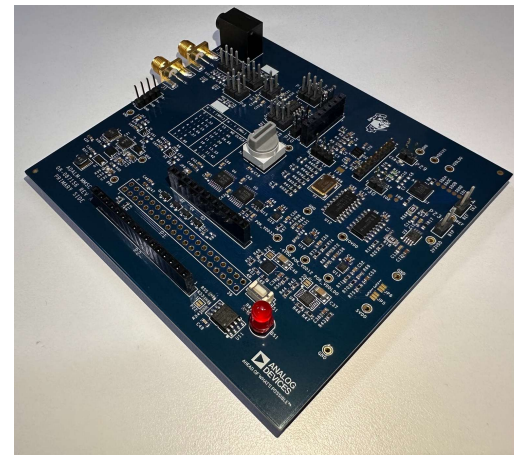
t.ly/ThG31

Connection Setup

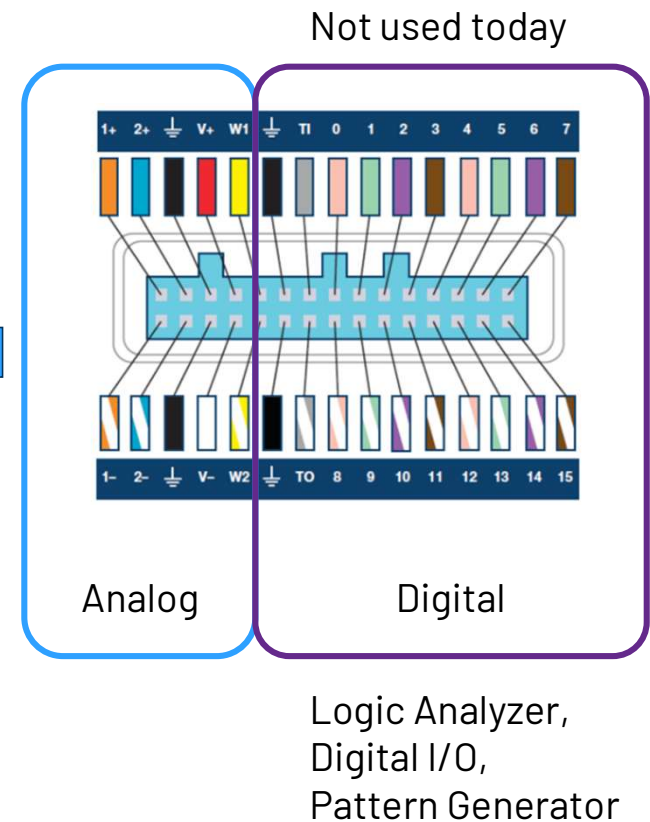
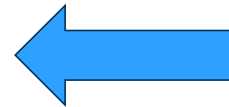
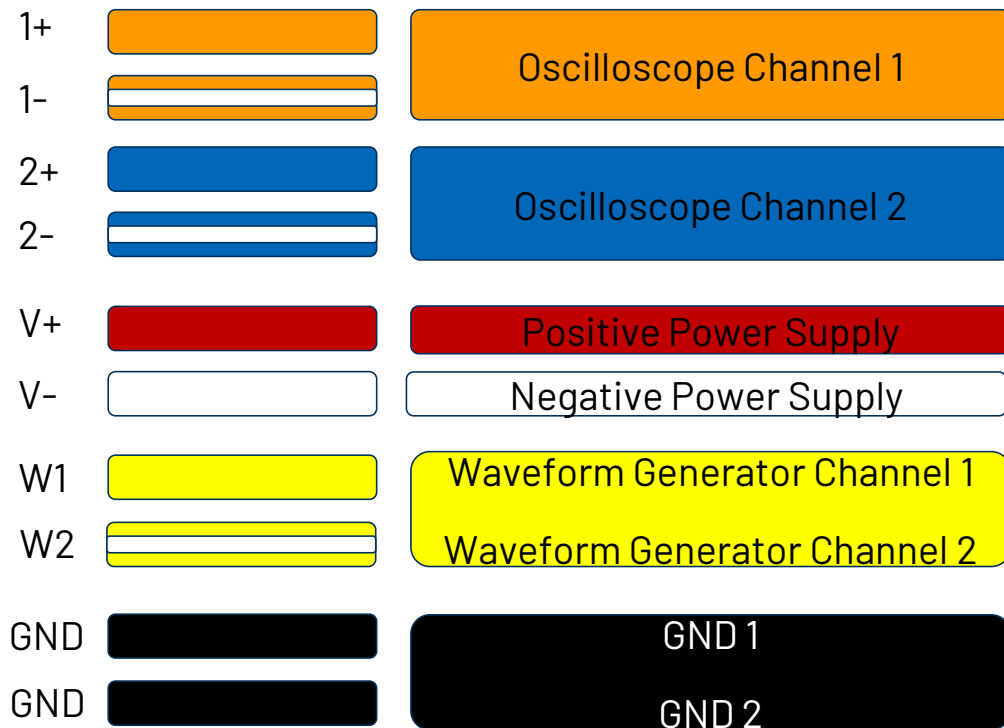
Hardware Setup

Equipment Used:

- ADALM-MMSC evaluation board
- ADALM2000 (M2K)
- 2x USB-micro to USB cables (provided)
- Laptop



Let's get familiar with the ADALM2000 Pinout



- Jumpers:

- P12, 1-2 (up)
- P10, 1-2 (down)
- P9, 1-2 (right)

(others are don't care for this experiment)



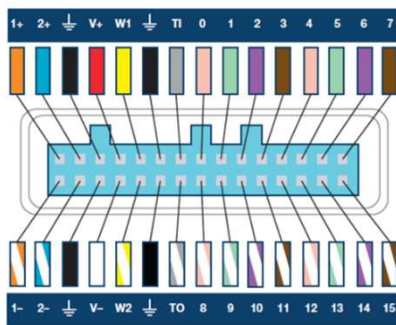
Connections Setup Photo Version (First Build Eratta!)

- W1 to IN+ on P17
- W2 to IN- on P17
- 1+ to P29, pin 2
- 1- to m2k GND
- 2+ to P30, pin 2
- 2- to m2k GND
- GND to P17 GND

Jumpers:

- P12, 1-2 (up)
- P10, 1-2 (down)
- P9, 1-2 (right)

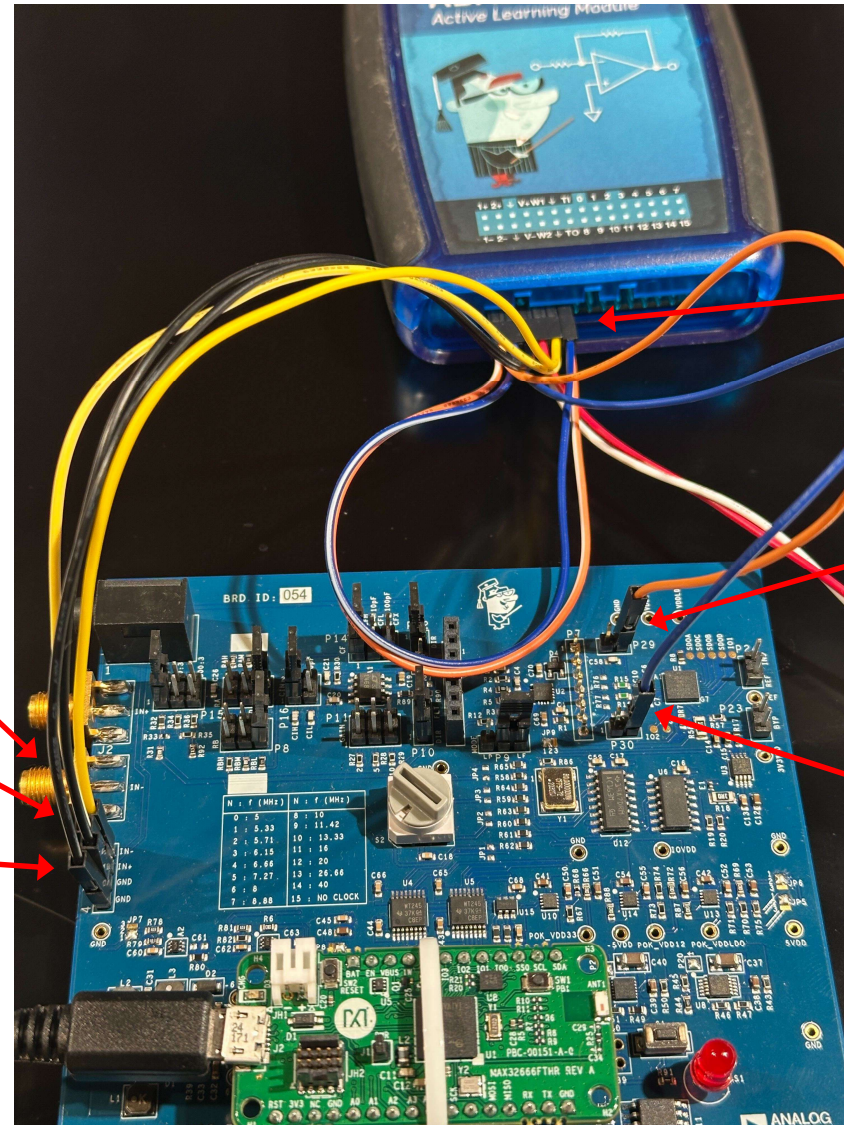
(others are don't care for this experiment)



M2k W1

M2k W2

GND (2X)



M2k 1- M2k 2-

1-, 2- looped back to next two pins, which are GND.

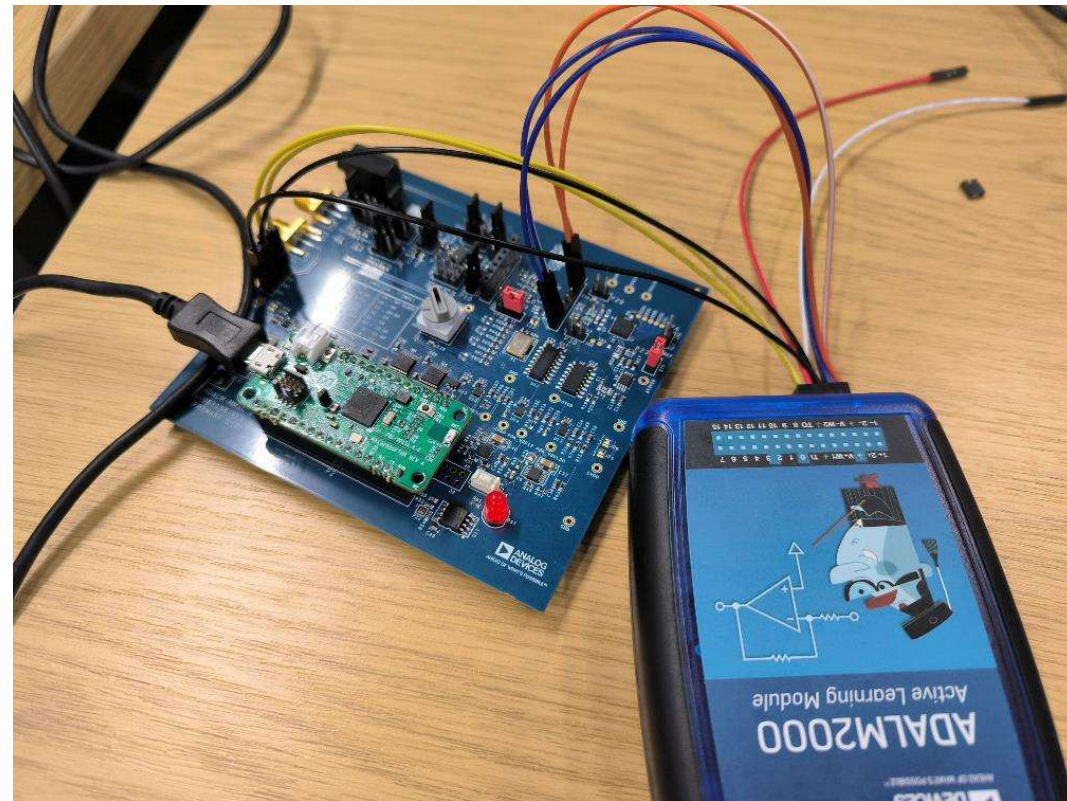
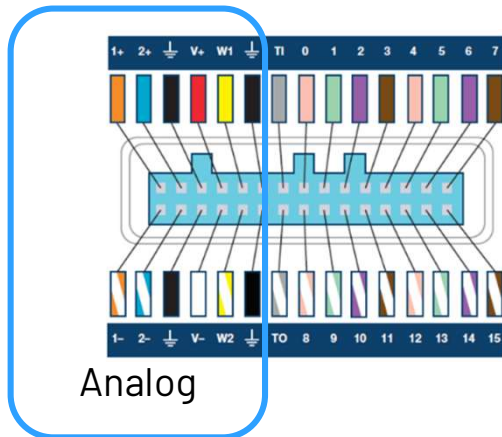
M2k 1+

M2k 2+



Hardware Power Up Sequence

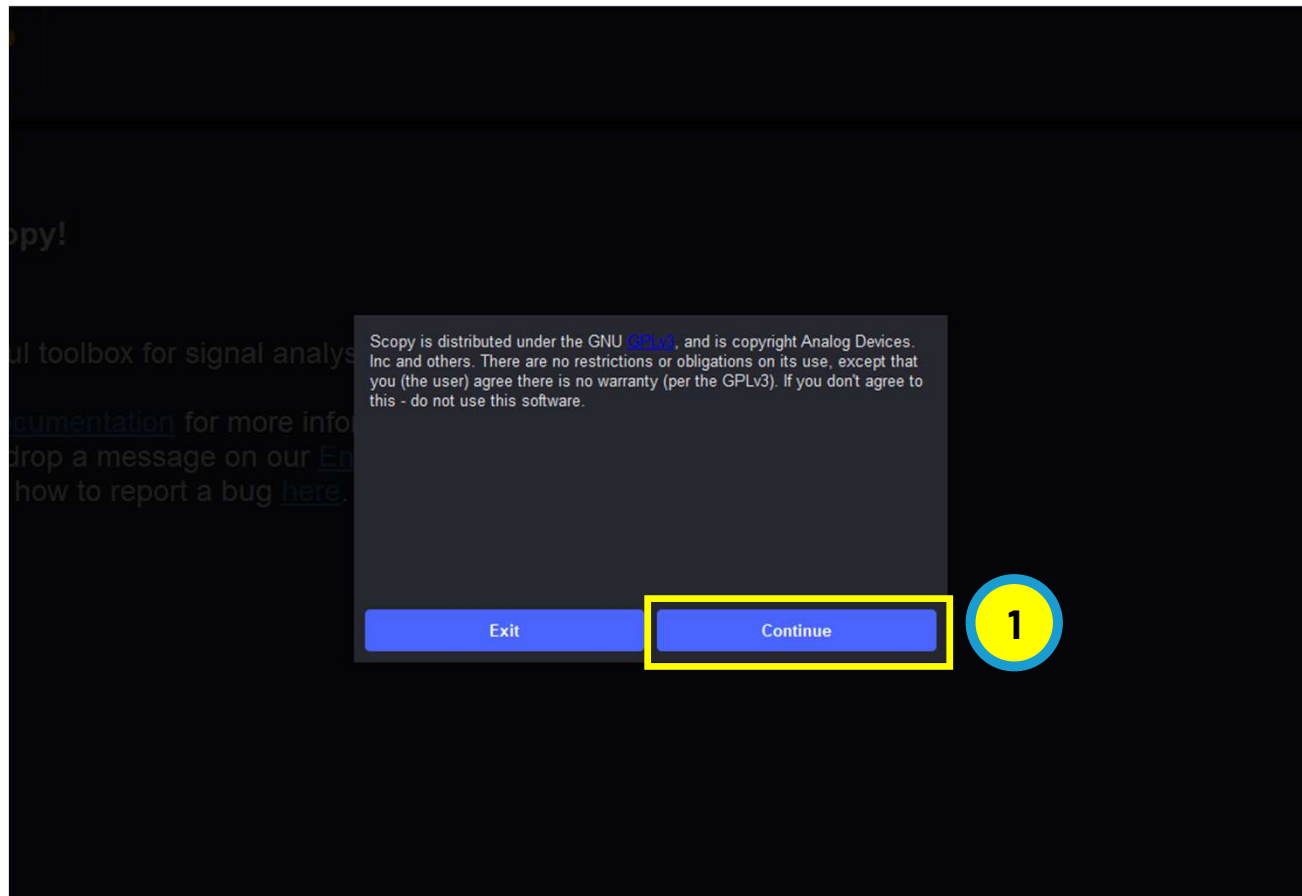
1. Plug Analog cable of the M2K only
2. Connect micro-USB from MAX32666FTHR to Laptop.
3. Connect micro-USB from ADALM2000 to Laptop.
 1. ADALM2000 has two USB ports – use the **CENTER** one





Scopy 2.2.0 & ADC plugin (IIO)

Launch Scopy



Scopy Software and Firmware Version Verification



1

Scopy - v2.2.0 - 50f66a373

Scopy - v2.2.0

2 Click on M2k Icon

3

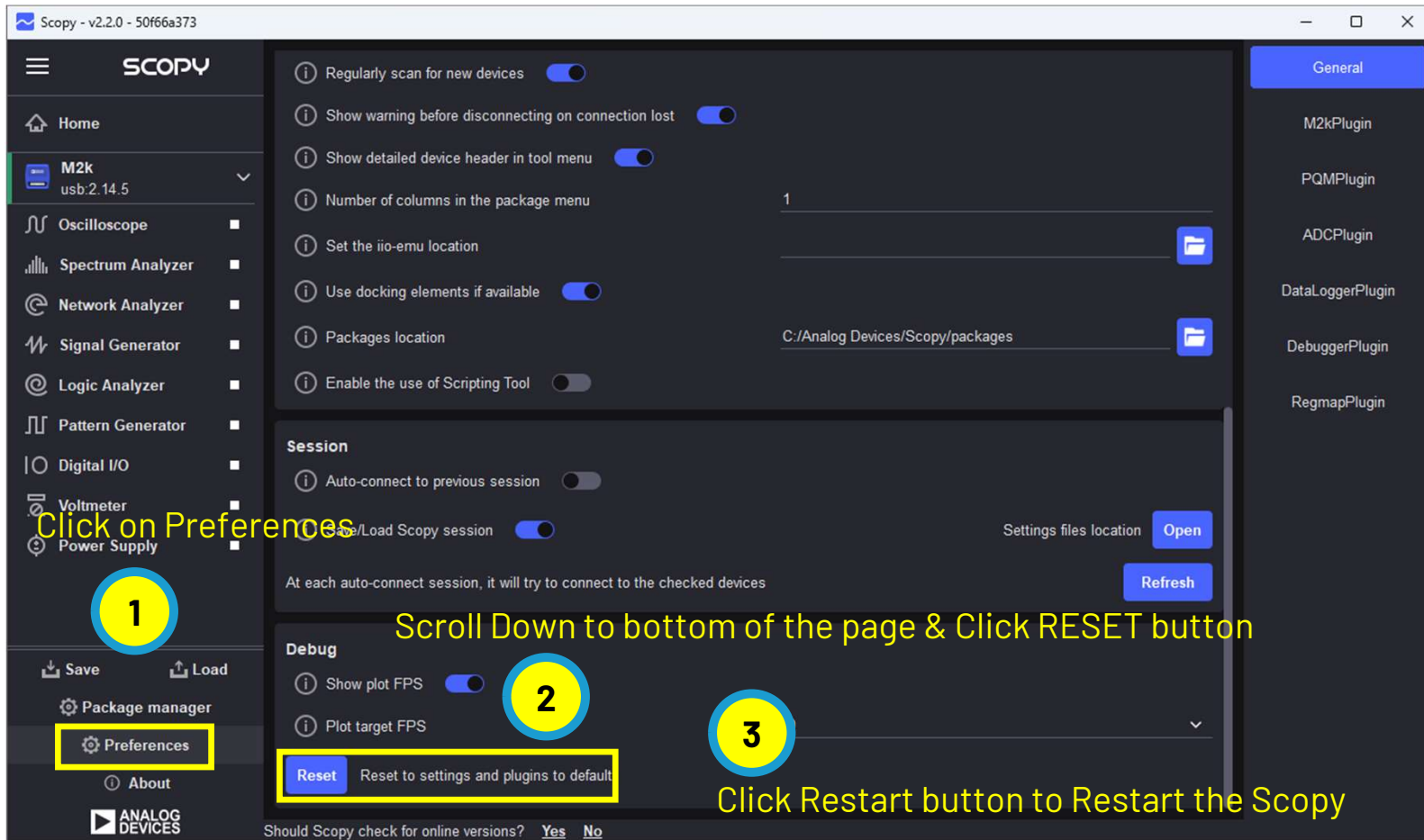
Check Firmware version = v0.33

The screenshot shows the Scopy software interface. On the left is a sidebar with navigation options: Home, M2k (usb:2.14.5), Oscilloscope, Spectrum Analyzer, Network Analyzer, Signal Generator, Logic Analyzer, Pattern Generator, Digital I/O, Voltmeter, and Power Supply. Below these are buttons for Save, Load, Package manager, Preferences, and About. The main area displays the M2k device icon (labeled M2k and usb:2.14.5) with a yellow box around it and a callout '2 Click on M2k Icon'. Below the icon are buttons for Connect, Identify, and Calibrate. A yellow box around the 'Firmware version' field shows 'v0.33' with a callout '3 Check Firmware version = v0.33'. Below this is a table of device information:

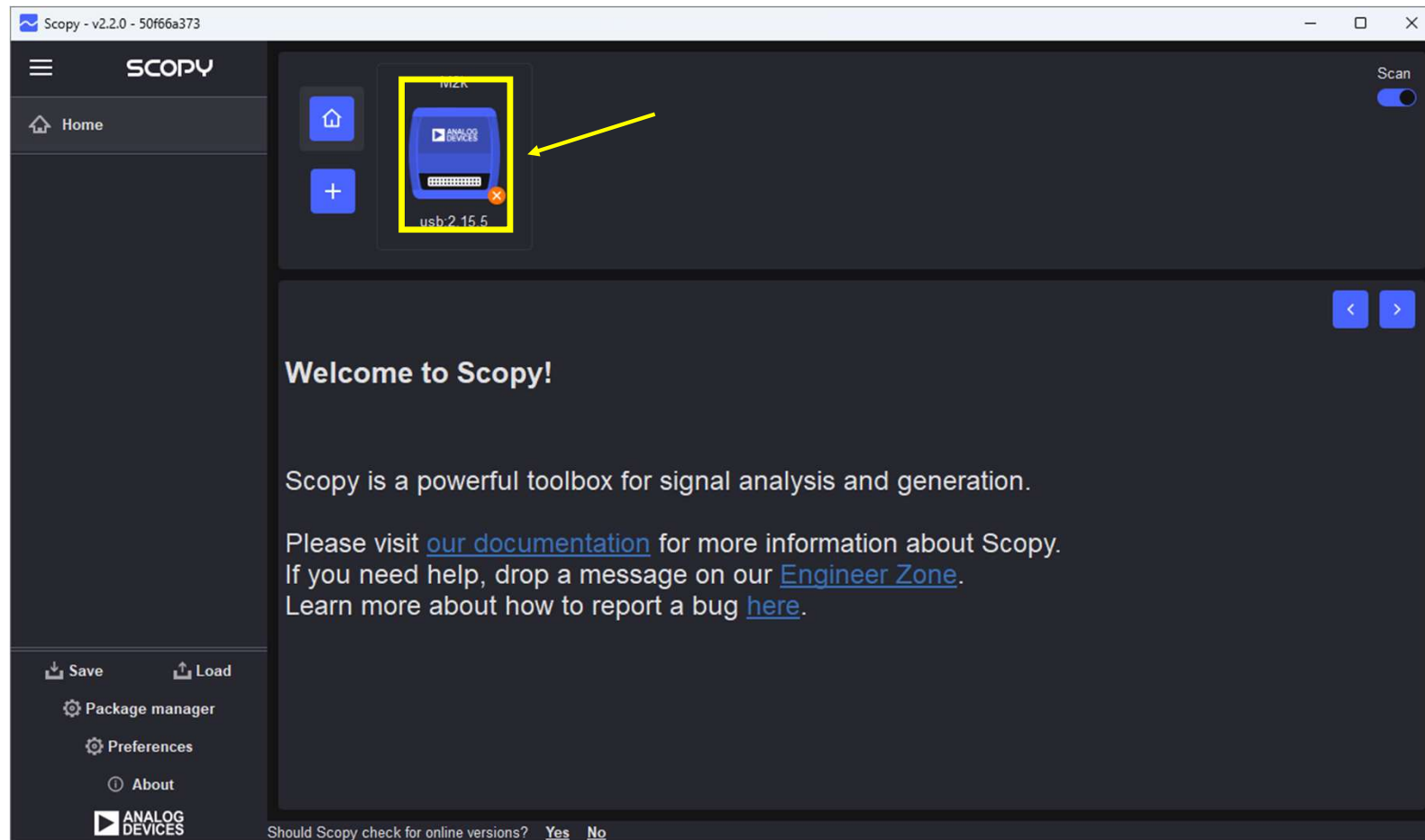
Model	Analog Devices M2k Rev.D (Z7010)
Model variant	1
Serial	1044734a420b0012ebff2400fa8a679607
Firmware version	v0.33
Kernel version	5.15.36-182064-g7aa63204d396
Vendor ID	0456
Product ID	b672
Release	2.0
Vendor	Analog Devices Inc.
Product name	M2k (ADALM-2000)

At the bottom, there are tabs for 'Device info' and 'Available plugins', and a checkbox 'Should Scopy check for online versions?' with 'Yes' and 'No' options.

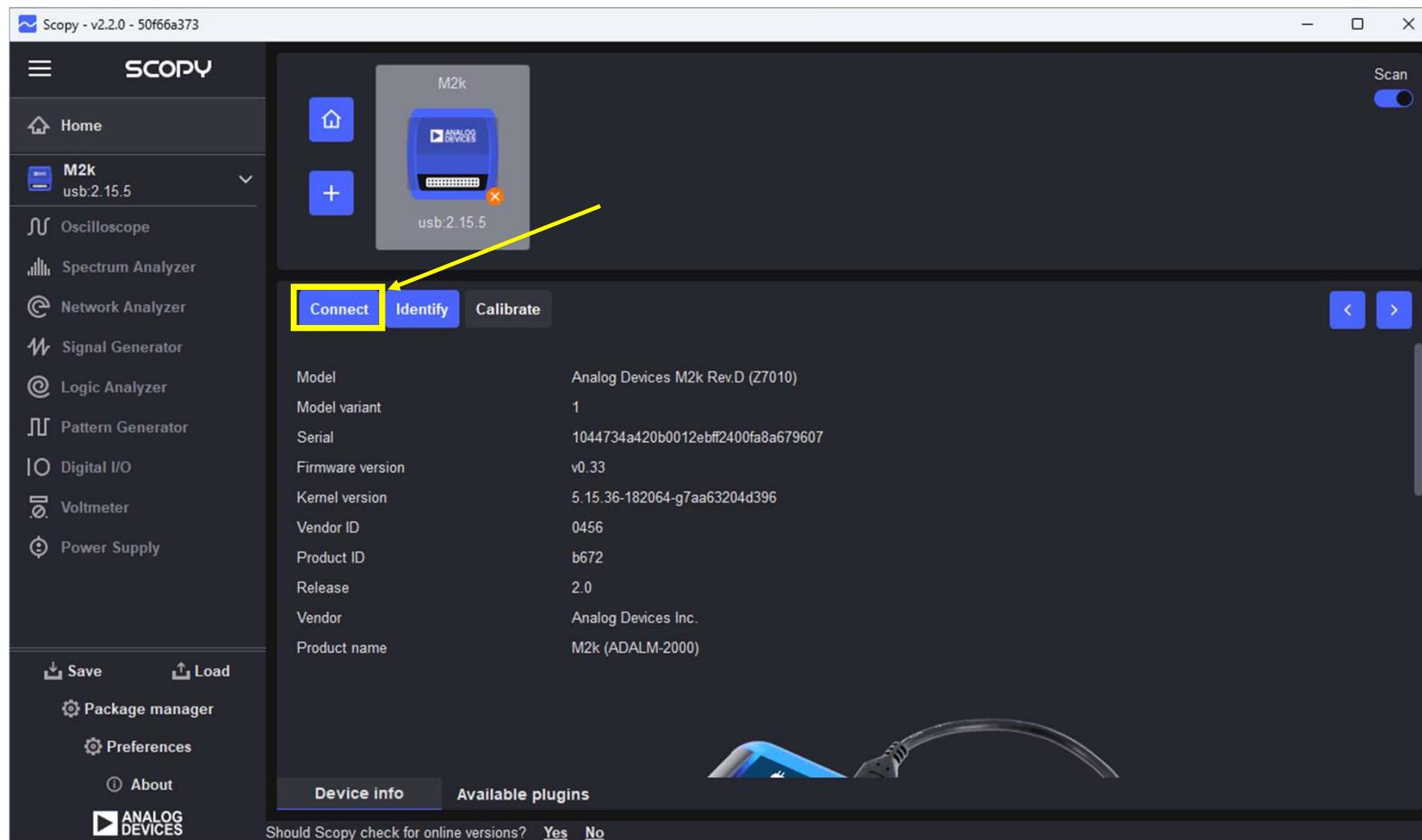
Reset to settings and plugins default



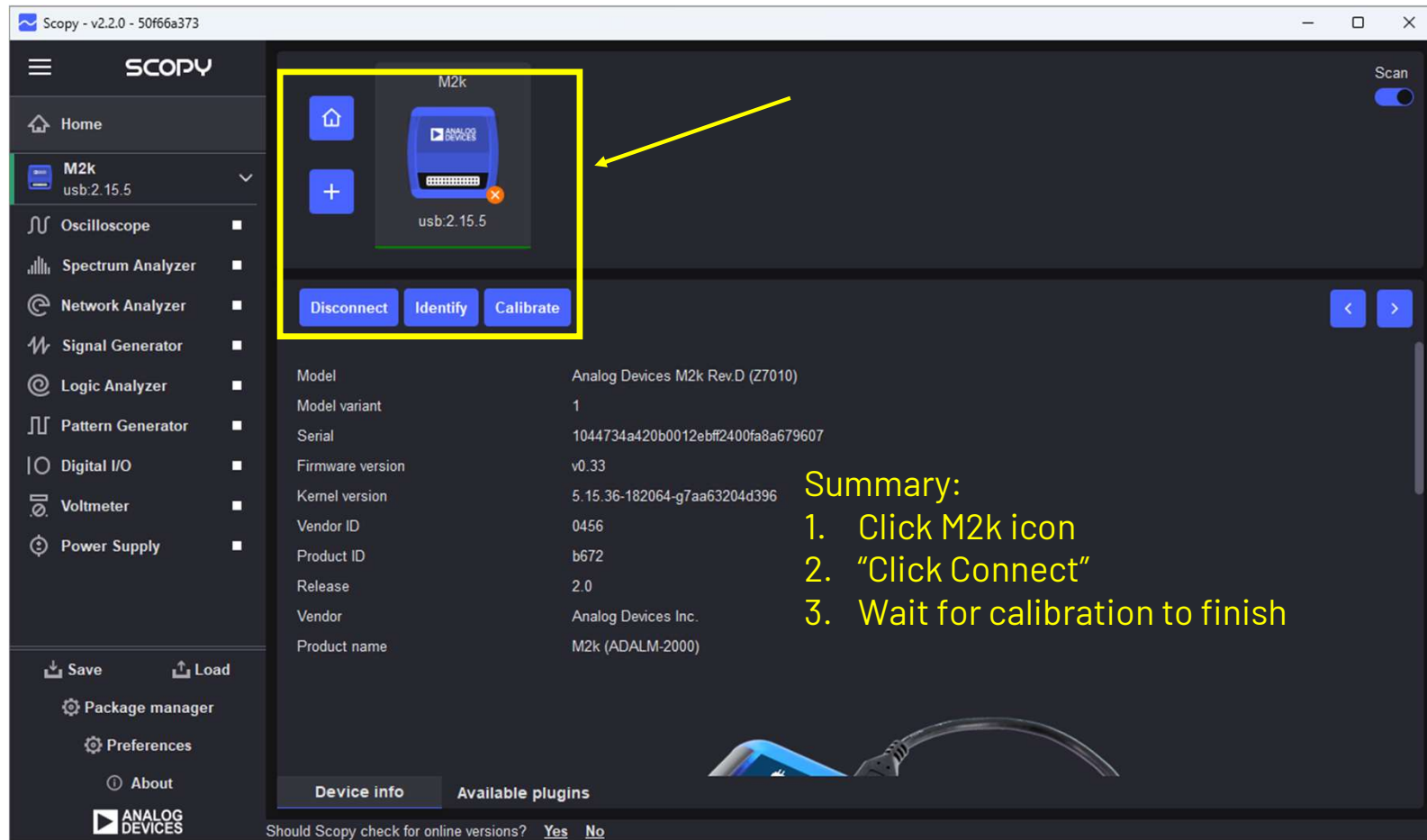
Click on the “M2k” Icon



Click "Connect"



Successful Connection Looks Like This



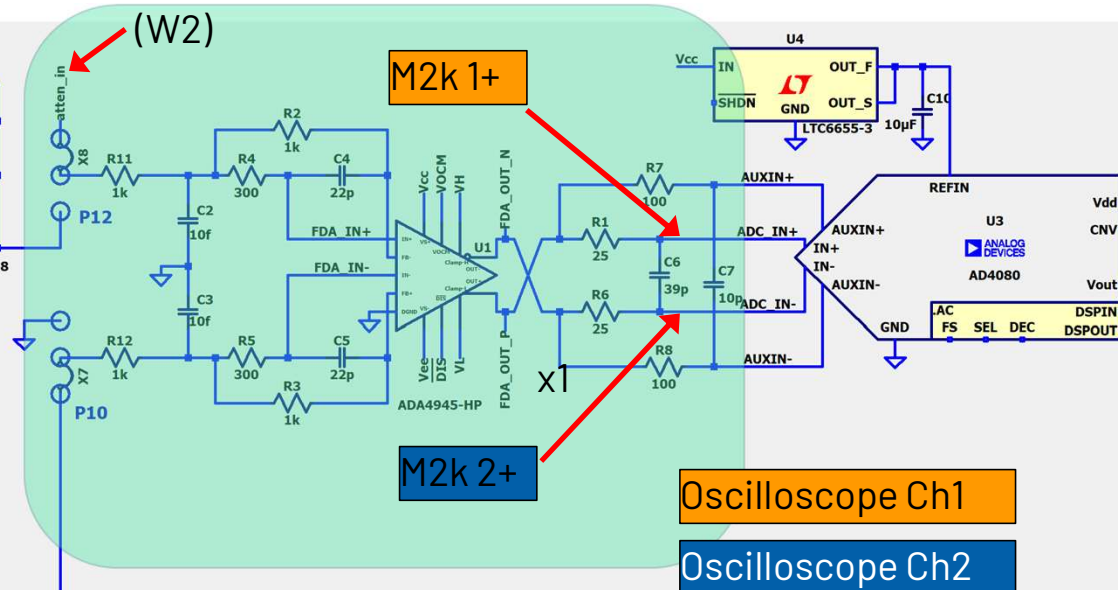
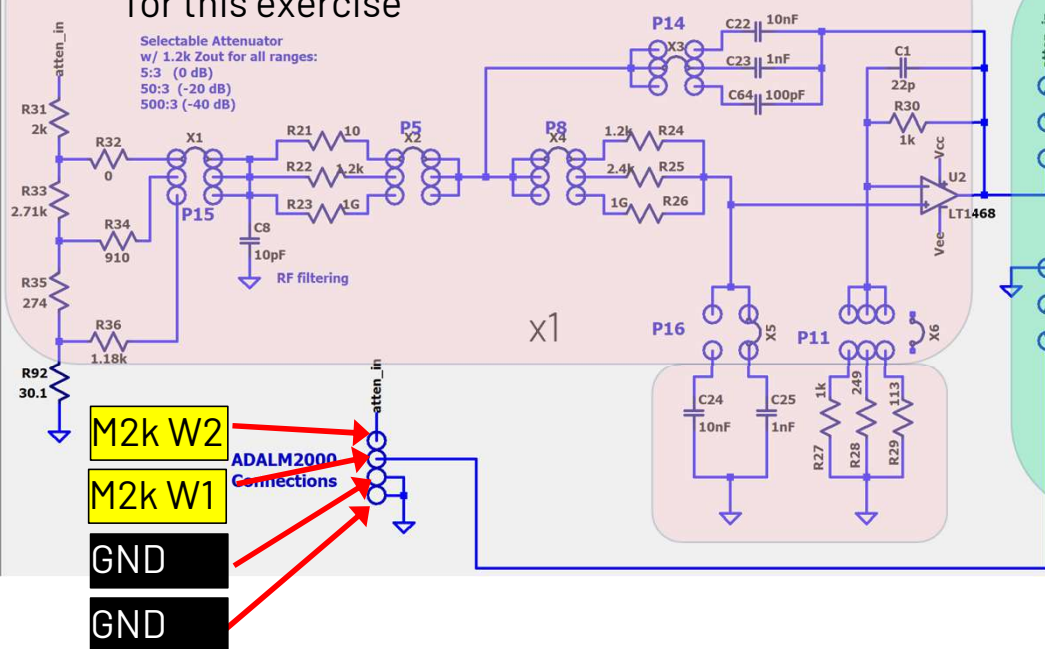
Print Me!

Signal Chain Schematic and measurement setup



Sallen Key filter is bypassed for this exercise

Selectable Attenuator
w/ 1.2k Zout for all ranges:
5:3 (0 dB)
50:3 (-20 dB)
500:3 (-40 dB)



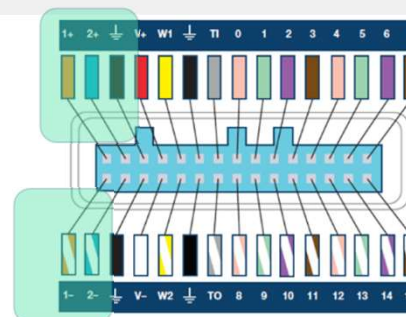
M2k W2

M2k W1

GND

GND

Waveform Generator



M2k 2-

M2k 1-



Signal Chain Signaling and FDA Function

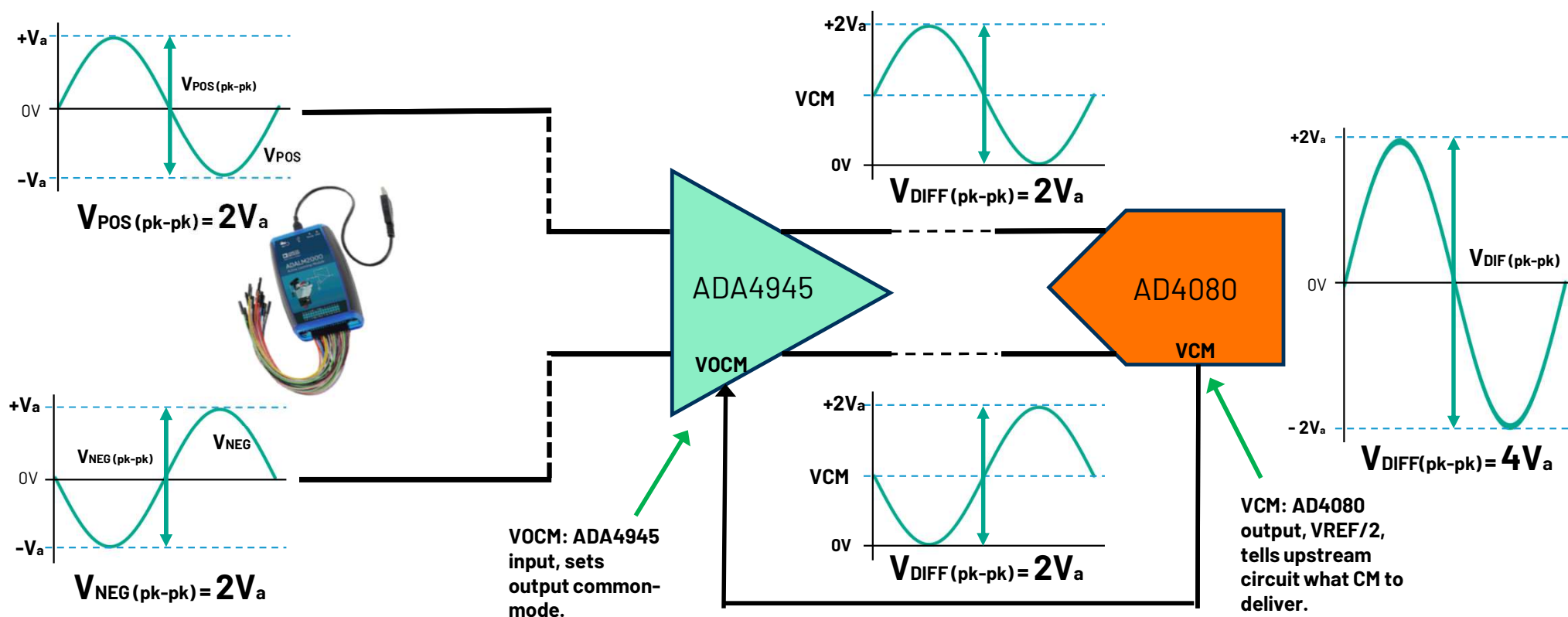


Gain Stage ($A_v=1$)

Buffer (ADC Driver) Stage ($A_v=1$)

ADC Stage

Precise Common-Mode Control



Visit Wiki Page

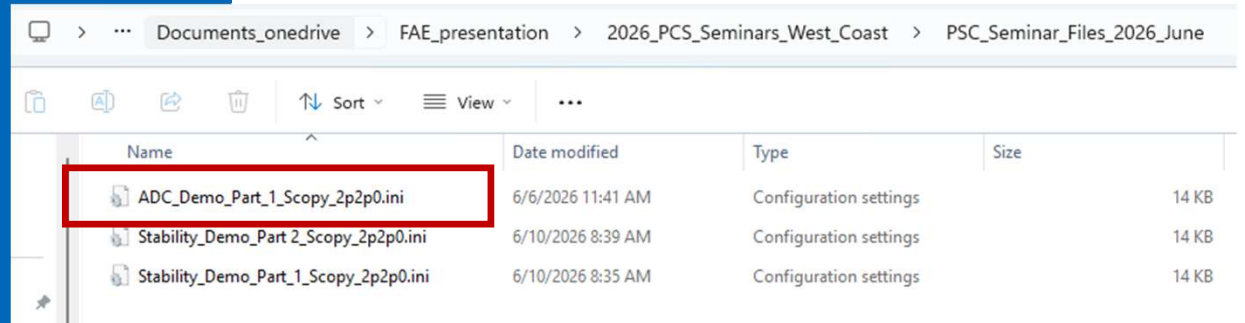


t.ly/ThG31

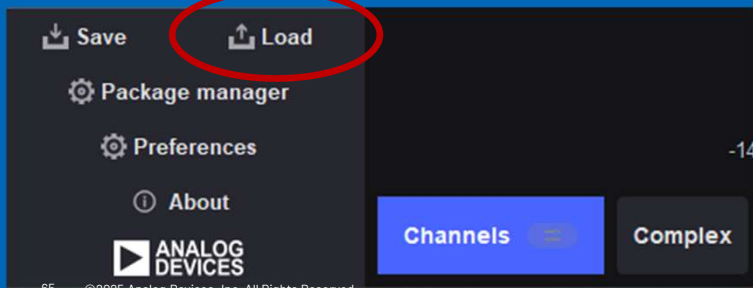
Part 1

Measuring Differential Signal

Load ADC_Demo_Part_1_Scopy_2p2p0.ini



Name	Date modified	Type	Size
ADC_Demo_Part_1_Scopy_2p2p0.ini	6/6/2026 11:41 AM	Configuration settings	14 KB
Stability_Demo_Part_2_Scopy_2p2p0.ini	6/10/2026 8:39 AM	Configuration settings	14 KB
Stability_Demo_Part_1_Scopy_2p2p0.ini	6/10/2026 8:35 AM	Configuration settings	14 KB



Signal Generator Channel 1 Verification



DO NOT CLICK on Run yet!

Signal Generator

1

2 CH1 - Left double-click

3 Waveform

4

Sine

⚡ 1 Vpp

625 kHz

Run

Constant Waveform Buffer Math

Sine

Amplitude 1 Volts p-p

Frequency 625 kHz

Offset 0 μ Volts

Phase 0 deg

NOISE

None

OTHER

Amplitude 1 μ Volts

Load (Ohm) inf

Line thickness 1

Plot Autoscale On Off

CH1 CH2

Should Scopy check for online versions? Yes No

Signal Generator Channel 2 Verification



Scopy - v2.2.0 - 50f66a373 - C:/Users/mthoren/OneDrive - Analog Devices, Inc/Documents_onedrive/FAE_presentation/2026_PCS_Seminars_West_Coast/PSC_Seminar_Files/Seminar_Files/ADC_Demo_Part...

SCOPE

Home

M2k
usb:2.14.5

Oscilloscope

Spectrum Analyzer

Network Analyzer

Signal Generator

Logic Analyzer

Pattern Generator

Digital I/O

Voltmeter

Power Supply

Save Load

Package manager

Preferences

About

ANALOG DEVICES

CH 1 CH 2

Should Scopy check for online versions? Yes No

OpenGL rendering
instrument: 0.092982 fps / 10754.8 ms

Constant Waveform Buffer Math

Rising Ramp Sawtooth

Amplitude
3
Volts p-p

Frequency
12.5
kHz

Offset
0
µVolts

Phase
0
deg

NOISE
None

OTHER
Load (Ohm) inf
Line thickness 1
Plot Autoscale On Off

Run

Click Run

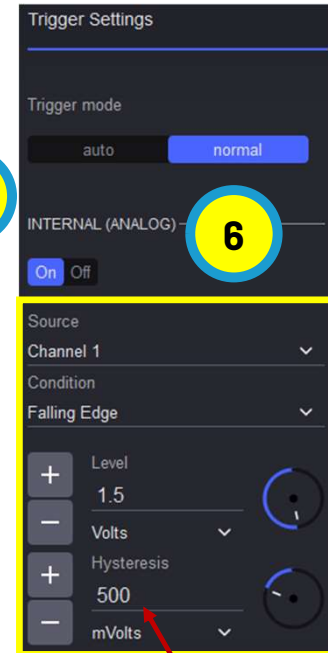
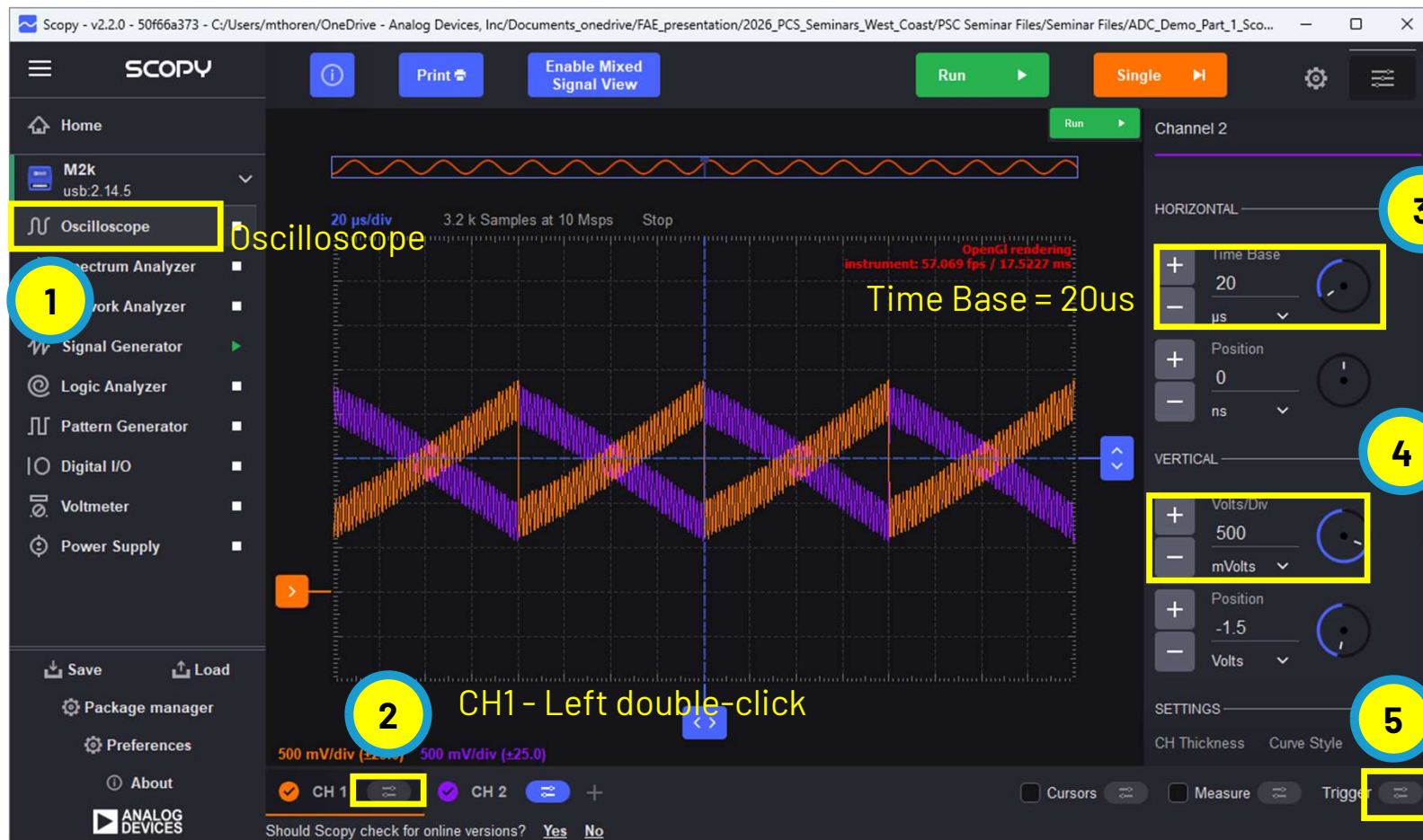
Rising Ramp Sawtooth

3 Vpp

12.5 kHz

CH2 - Left double-click

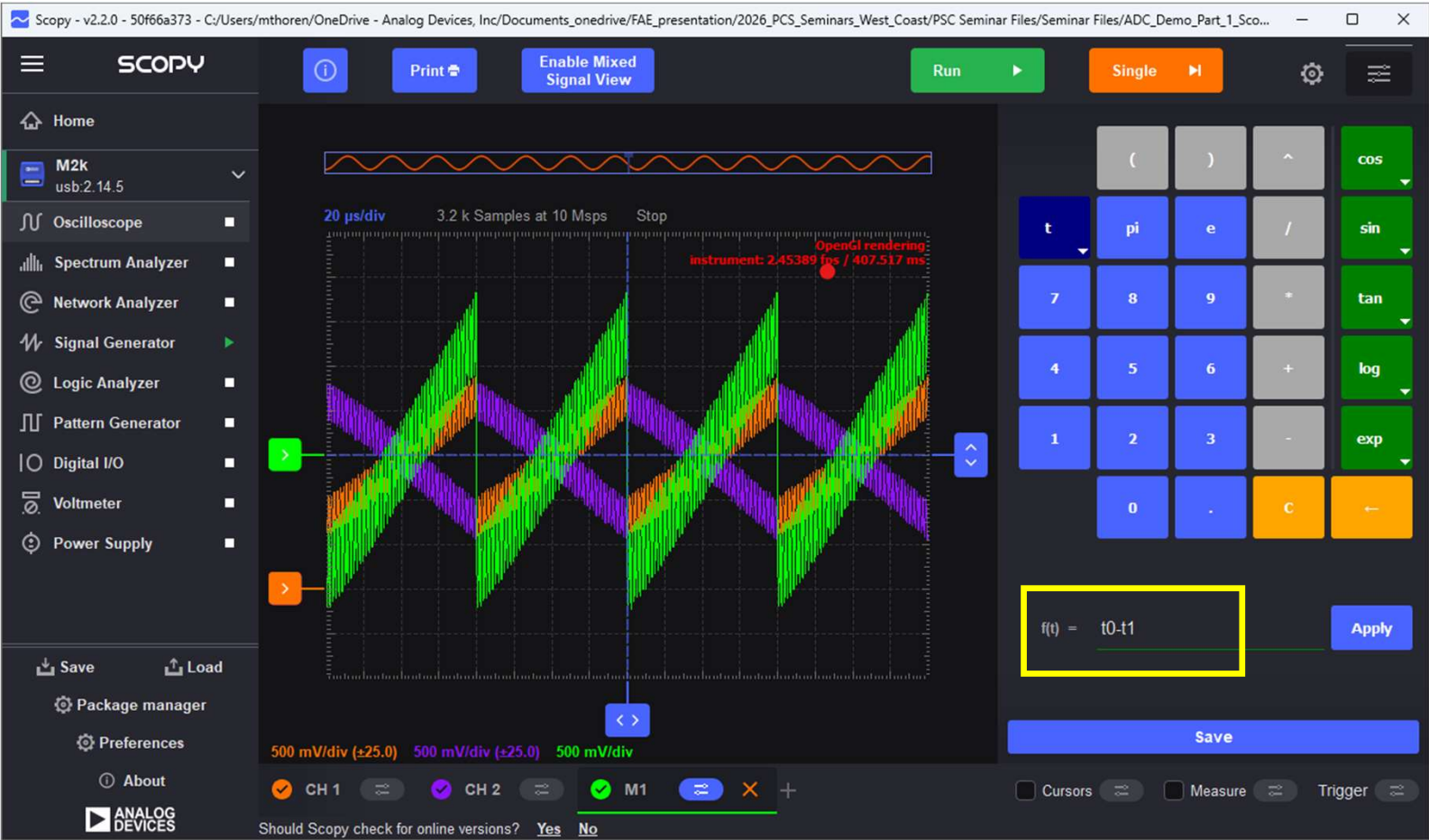
Oscilloscope Ch 1 – probing differential signals wrt. GND



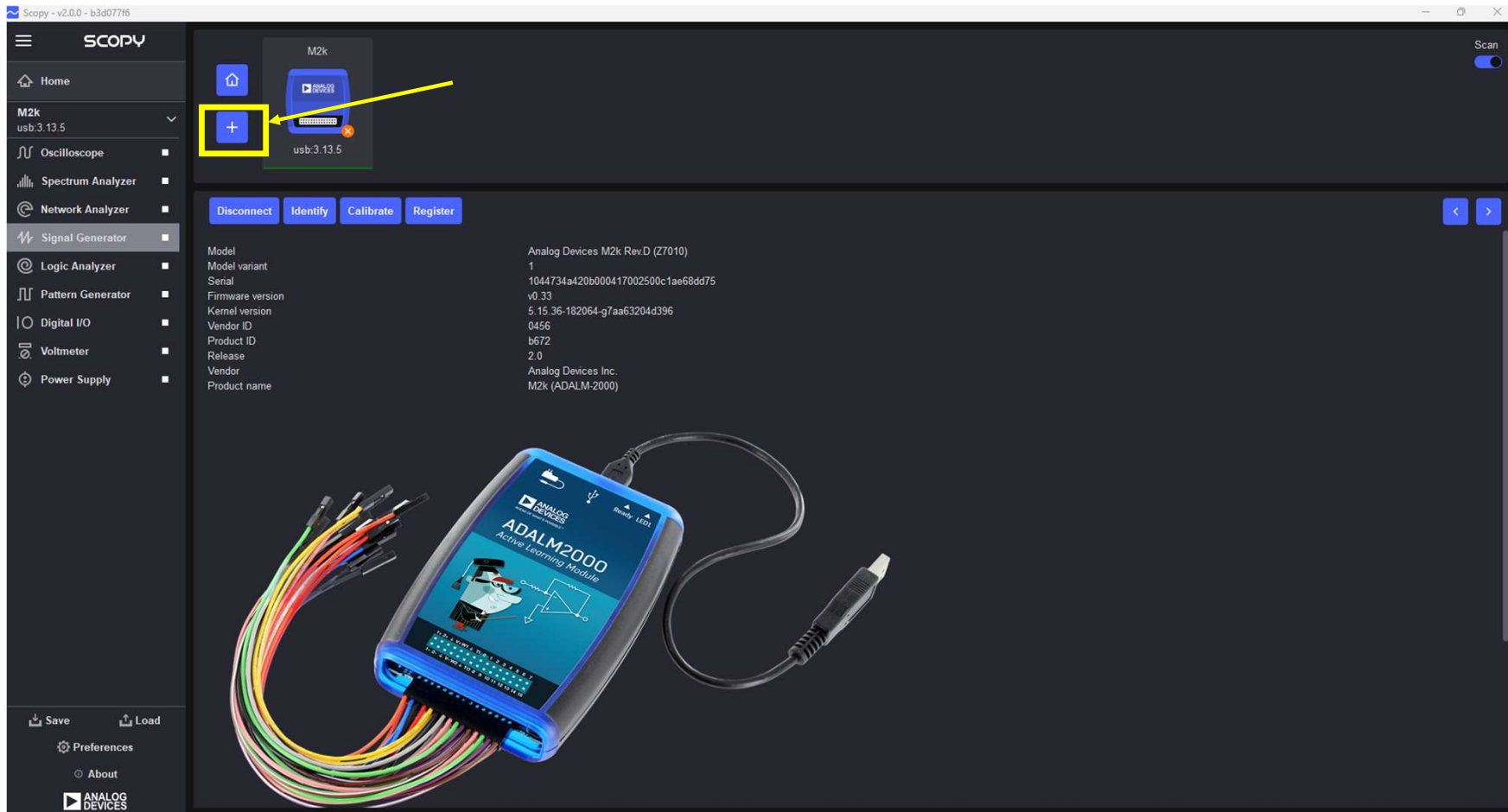
500 mV hysteresis prevents "jumping"



Math: CH1 – CH2 is what the AD4080 responds to

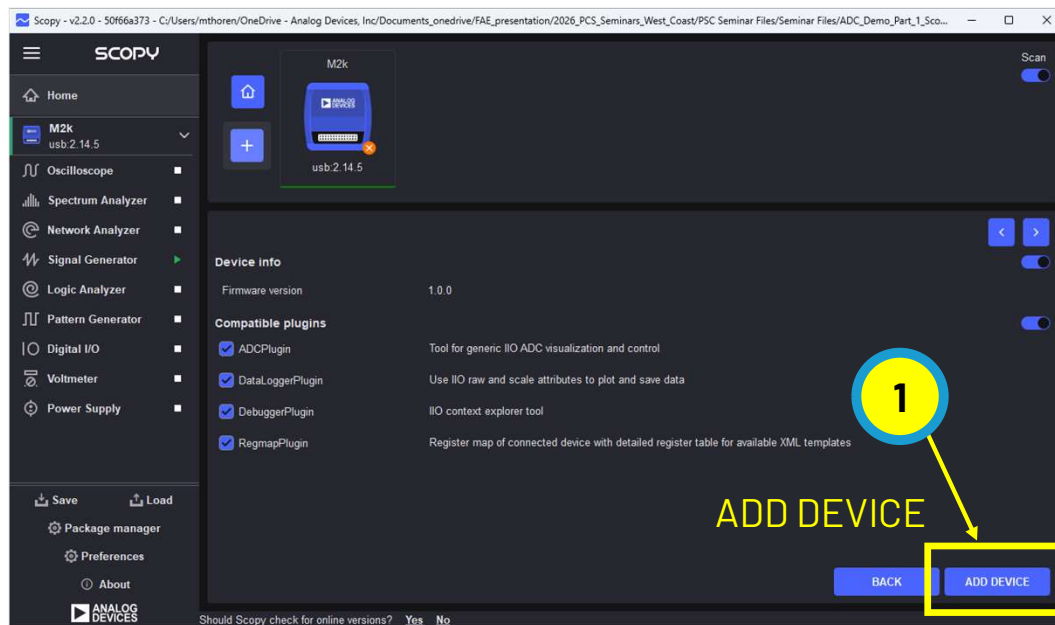


Adding IIO Connection

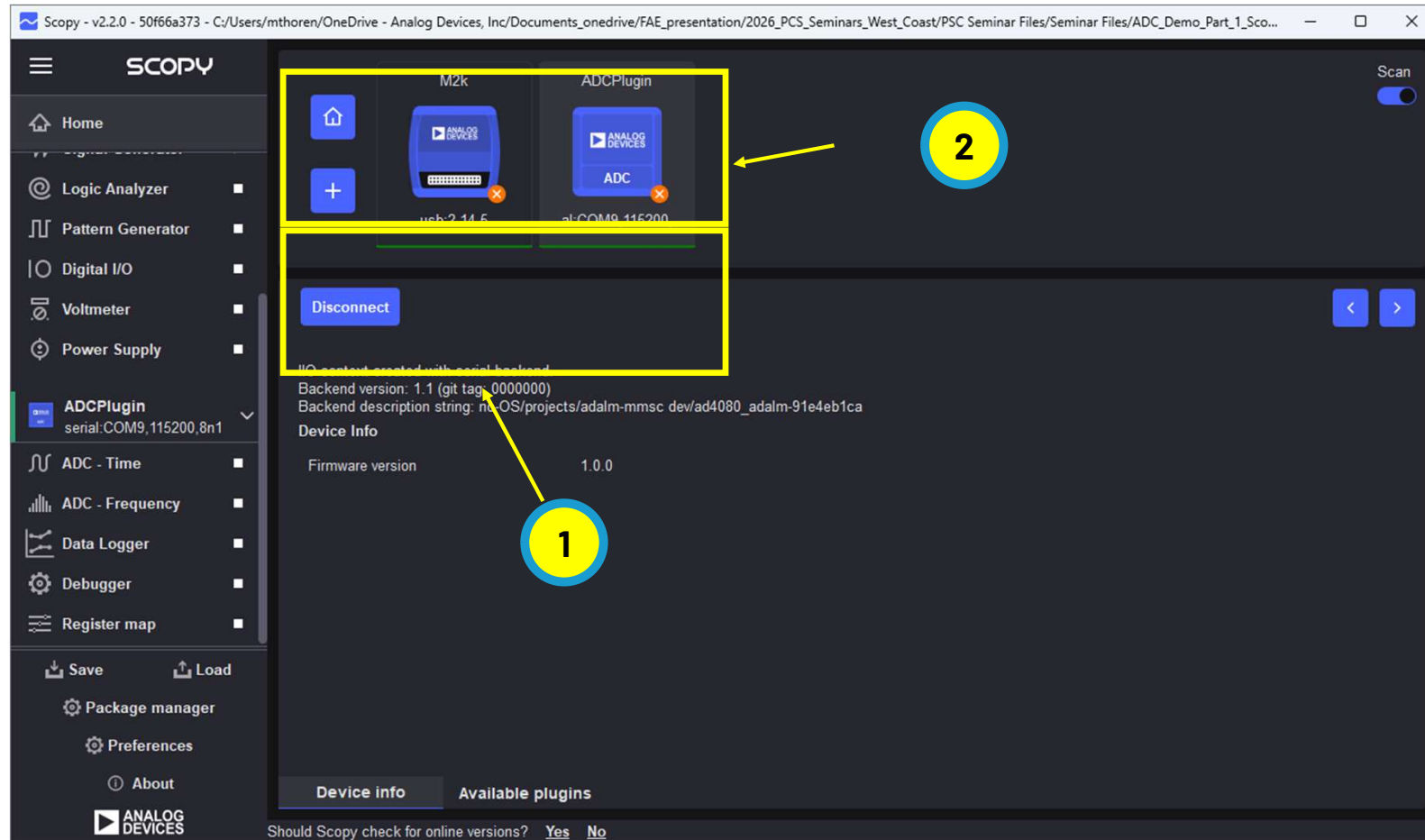




Add Device and Connect

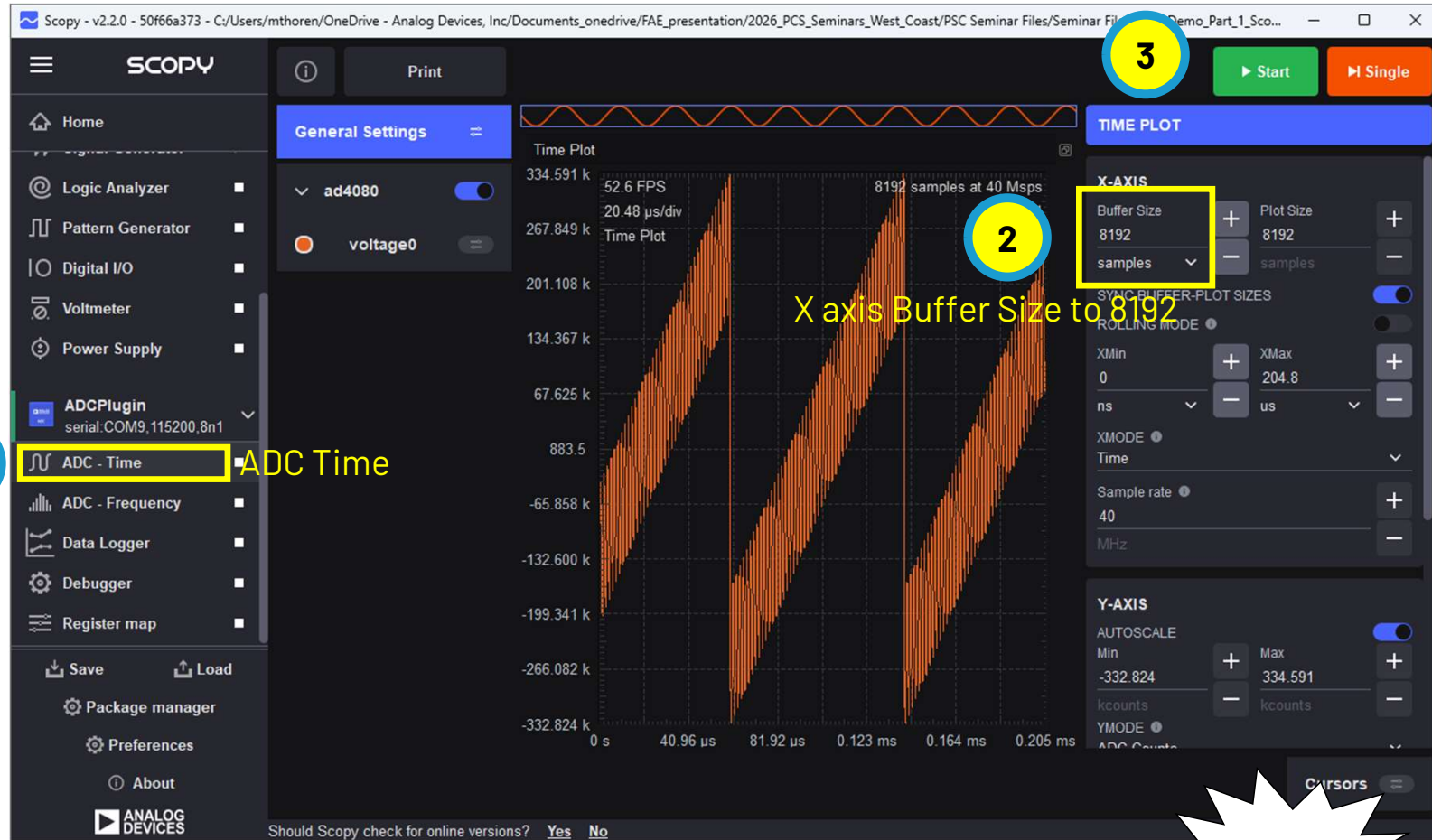


Successful Connection Looks Like This



ADC Data!

1. Go to ADC Time
2. Change buffer size in X-Axis to:
1024 (1.024ksamples)
3. Click Start
4. Observe waveform
5. Click Stop



Visit Wiki Page



t.ly/ThG31

Part 2

Setting up Digital Filter

Digital Filter Characteristics – Sinc1

$f_{\text{sample}} = 40 \text{ MHz}$

If $\text{dec_rate} = 8 \rightarrow$

$\text{ODR} = f_{\text{sample}} / \text{dec_rate}$

Notch at $k * F_s / \text{Dec rate} = k * 40\text{M} / 8 = k * 5\text{MHz}$
(where k is integer number)

Nyquist Bandwidth: 20MHz

Attenuation = Filter order * (-20dB)

Sinc1 = -20dB/dec

Sinc3 = -60dB/dec

Sinc5 = -100dB/dec

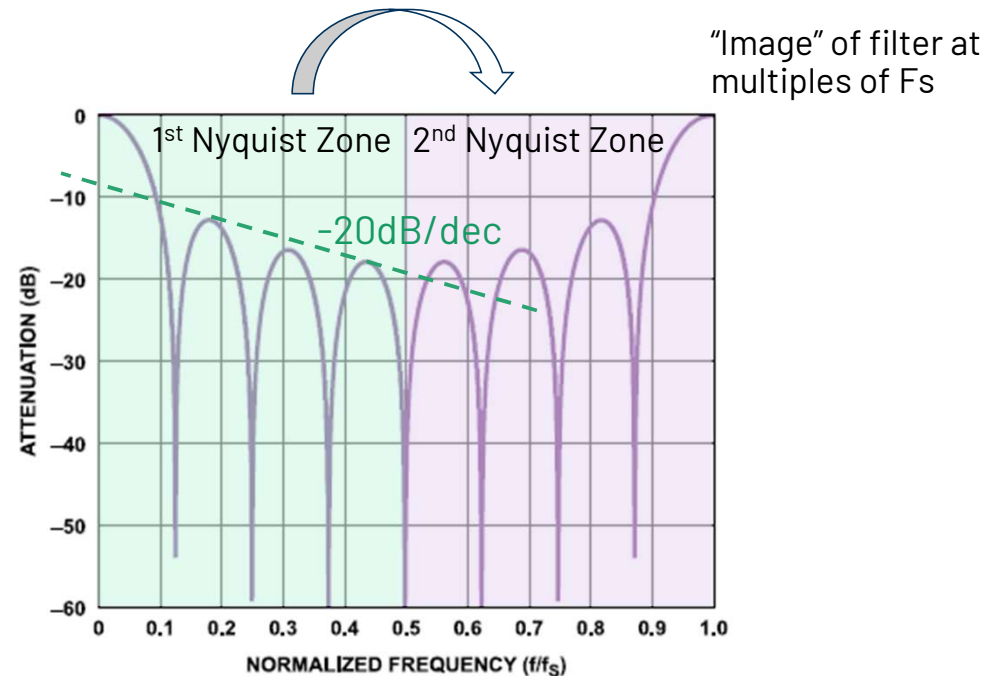
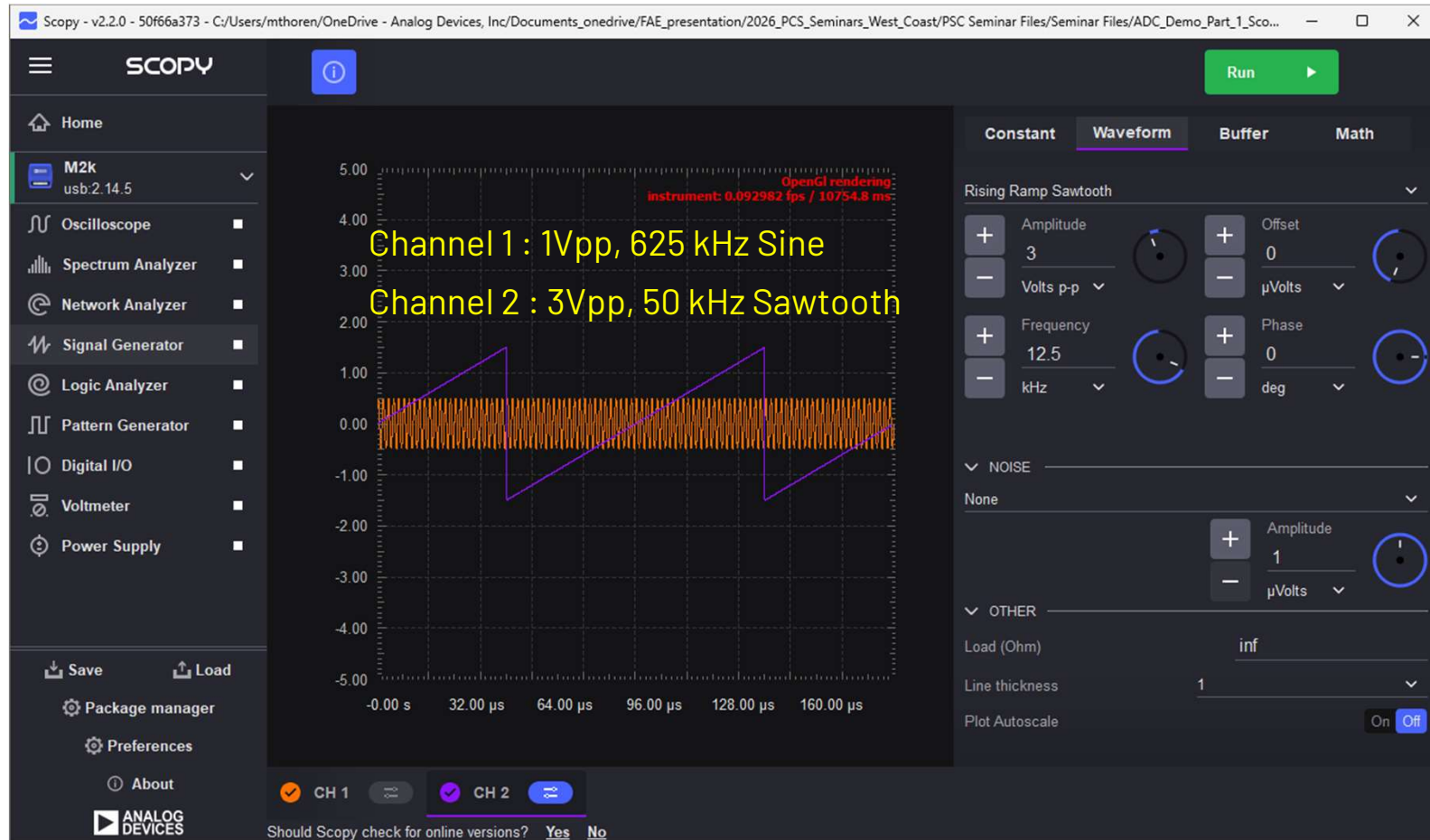


Figure 102. Sinc1 Filter Response, Decimate by 8

Nyquist Bandwidth = $F_s / 2$ (0.5 Normalized)

Signal Generator – Review Settings



Setting up Digital Filter type

1. Debugger:
(to access IIO attributes)
2. Expand iio:device0:
ad4080
3. Click on filter_sel
4. Select Sinc1
in the attributes window

The screenshot shows the Analog Devices GUI with the following elements:

- Sidebar:** Contains navigation options like Home, Oscilloscope, Spectrum Analyzer, Network Analyzer, Signal Generator, Logic Analyzer, Pattern Generator, Digital I/O, Voltmeter, Power Supply, ADCPlugin, ADC - Time, Frequency, Data, **Debugger** (highlighted with a yellow box and callout 1), Register map, Save, Load, Package manager, Preferences, and About.
- Tree View:** Displays a hierarchy of IIO attributes. The 'iio:device0: ad4080' node is expanded (callout 2). The 'filter_sel' attribute is highlighted with a yellow box and callout 3.
- Right Panel:** Shows the 'Attributes' section with a list of attributes. 'sinc1' is selected (callout 4). Below this, the 'General info' section provides details about the device.

Name	Value	Type	Path
filter_sel	sinc1		

Setting up Digital Filter Decimation Rate**



0. "oversampling_ratio" = decimation rate

1. Select oversampling_ratio

2. Set to: 64

Q: Where is the first notch?

The screenshot shows the SCOPE GUI interface. On the left is a sidebar menu with options like Home, Logic Analyzer, Pattern Generator, Digital I/O, Voltmeter, Power Supply, ADCPlugin (selected), ADC - Time, ADC - Frequency, Data Logger, Debugger, and Register map. The main area is divided into a search bar, a list of attributes, and a detailed view of the selected attribute. The attribute list on the left has 'oversampling_ratio' highlighted with a yellow box and a blue circle labeled '1'. The detailed view on the right shows the 'oversampling_ratio' attribute with a value of 64, highlighted with a blue circle labeled '2'. Below this, there is a table with columns Name, Value, Type, and Path, showing the 'oversampling_ratio' attribute.

Name	Value	Type	Path
oversampling_ratio	64		

**Right or wrong, terminology difference is due to Linux driver's precedent

Calculating ODR and notch frequencies (sinc only)

$f_{\text{sample}} = 40 \text{ MHz}$

filter_type = Sinc1

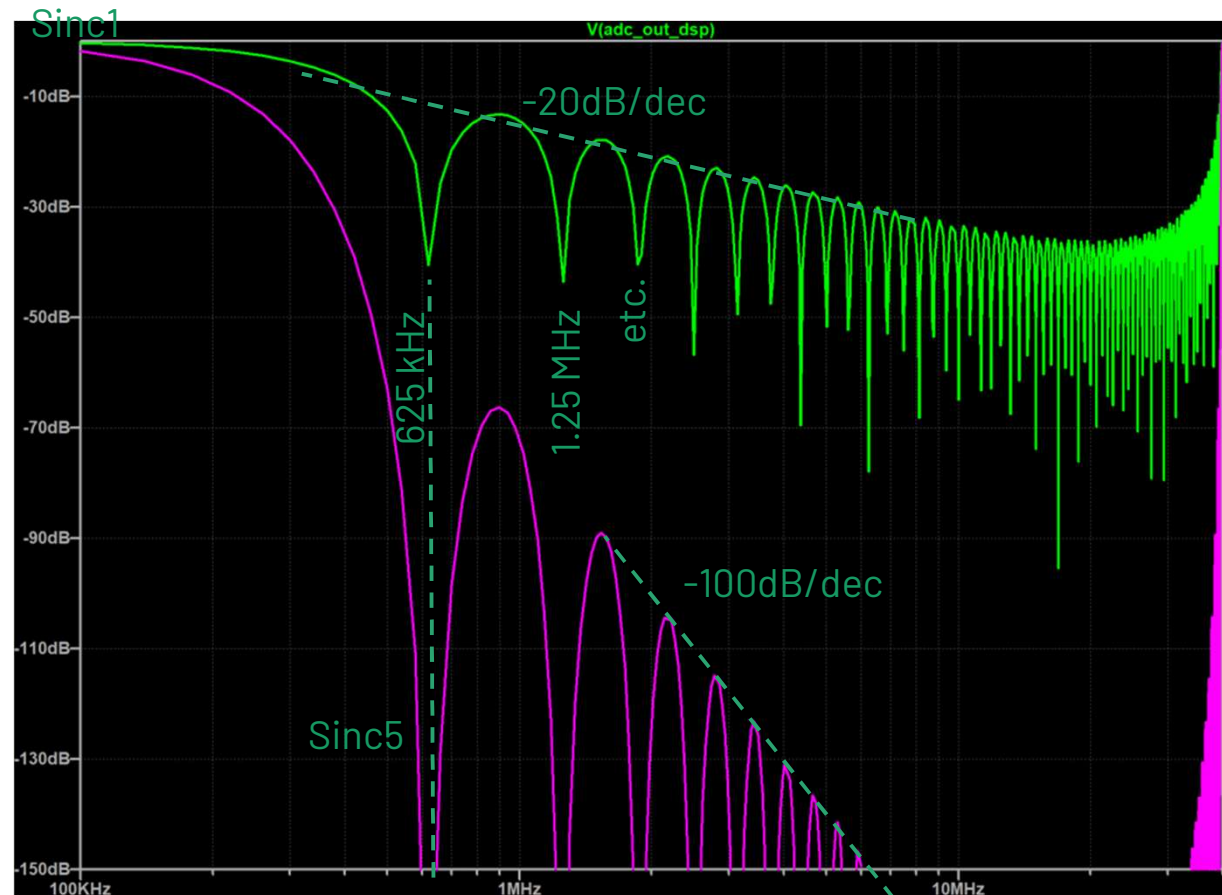
Oversampling_ratio = 64

(This means decimation rate = 64)

$\text{ODR} = f_s / \text{Decimation Rate} = 625 \text{ kHz}$

Notches at $k * f_s / \text{Dec rate} = k * 625 \text{ MHz}$

** Important! For a given front-end sample rate, increasing decimation reduces the output data rate. The same number of output samples represents a longer time interval.



ANALOG
DEVICES



The screenshot displays the Scopy v2.2.0 software interface. The top window title bar shows the file path: "C:\Users\mthoren\OneDrive - Analog Devices\FAE\presentation\2026_PCS_Seminars_West_Coast\PSC Seminar Files\Seminar Files\Part_1_Sco...". The main interface is divided into several sections:

- Left Sidebar:** Contains navigation icons and a list of modules: Home, Logic Analyzer, Pattern Generator, Digital I/O, Voltmeter, Power Supply, ADCPlugin (selected, serial: COM4, 115200, 8n1), ADC - Time, ADC - Frequency, Data Logger, Debugger, Register map, Save, Load, Package manager, Preferences, and About.
- Top Bar:** Includes a menu icon, the "SCOPY" logo, an information icon, a "Print" button, and "Start" and "Single" buttons.
- General Settings Panel:** Shows the selected device as "ad4080" and the input as "voltage0".
- Time Plot:** A large central plot area showing a high-frequency digital signal. The y-axis ranges from -267.747 k to 249.027 k. The x-axis shows time intervals: 0 s, 40.96 μ s, 81.92 μ s, 0.123 ms, 0.164 ms, and 0.205 ms. Text overlays on the plot indicate "52.6 FPS", "20.48 μ s/div", and "8192 samples at 40 Msps".
- TIME PLOT Panel:** Contains configuration options for the plot:
 - X-AXIS:** Buffer Size (8192), Plot Size (8192), samples, SYNC BUFFER-PLOT SIZES (checked), ROLLING MODE (unchecked), XMin (0 ns), XMax (204.8 μ s), XMODE (Time), Sample rate (40 MHz).
 - Y-AXIS:** AUTOSCALE (checked), Min (-267.747 kcounts), Max (249.027 kcounts), YMODE (ADC Counts).
- Bottom Right:** A "Cursors" button.

Digital Filter removes the sine wave

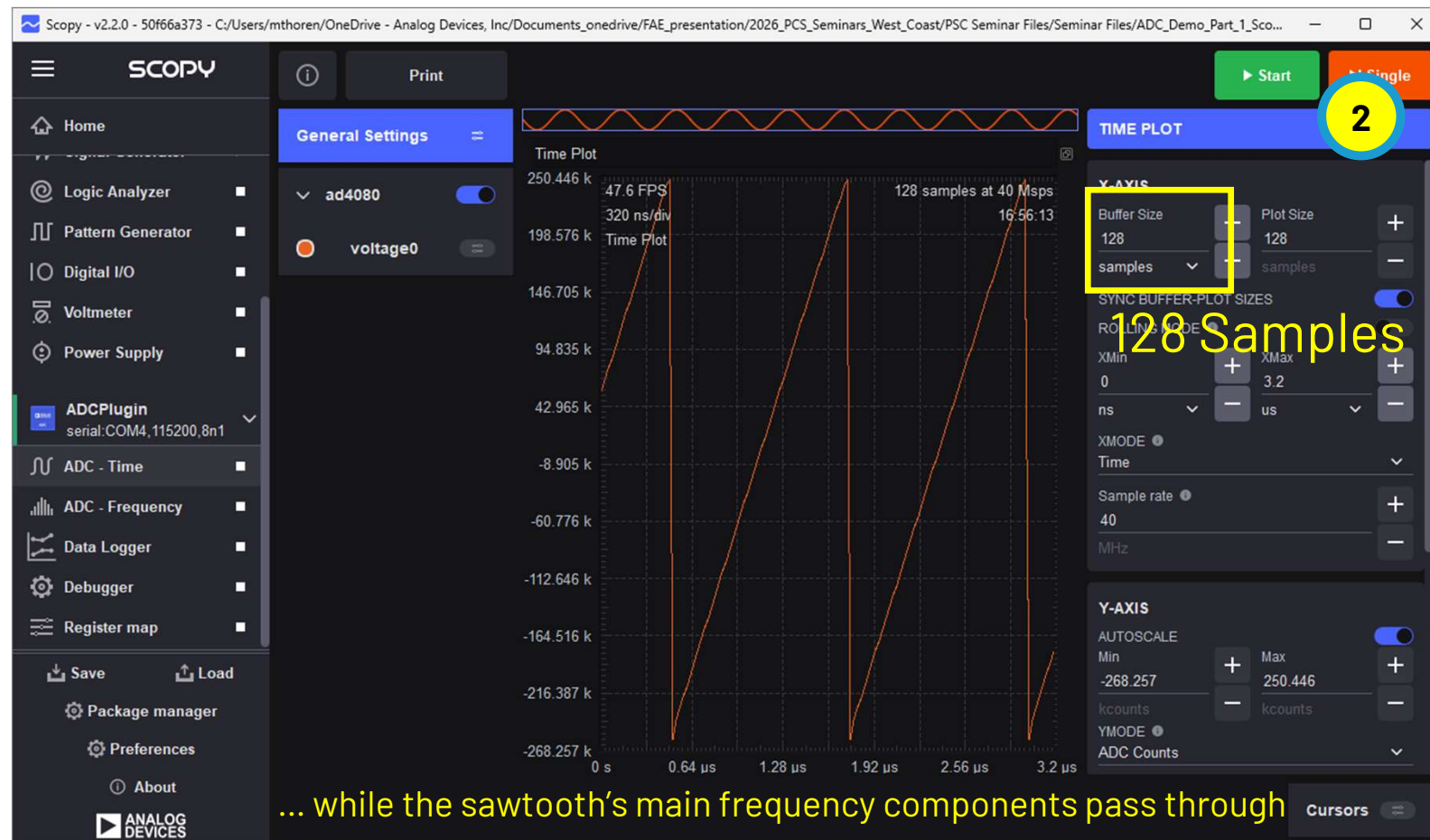


Click **Start**



That's a *lot* less data... but each point contains 64 samples averaged, so it's higher quality data, with the interferer strategically removed.

Also – that's less data the downstream processor has to deal with.



... while the sawtooth's main frequency components pass through

Try reducing DF to 32...



Consider:

$f_{\text{sample}} = 40 \text{ MHz}$

$\text{dec_rate} = 32,$

$-3 \text{ dB} @ 0.443 * f_{\text{sample}} / R$

541 kHz

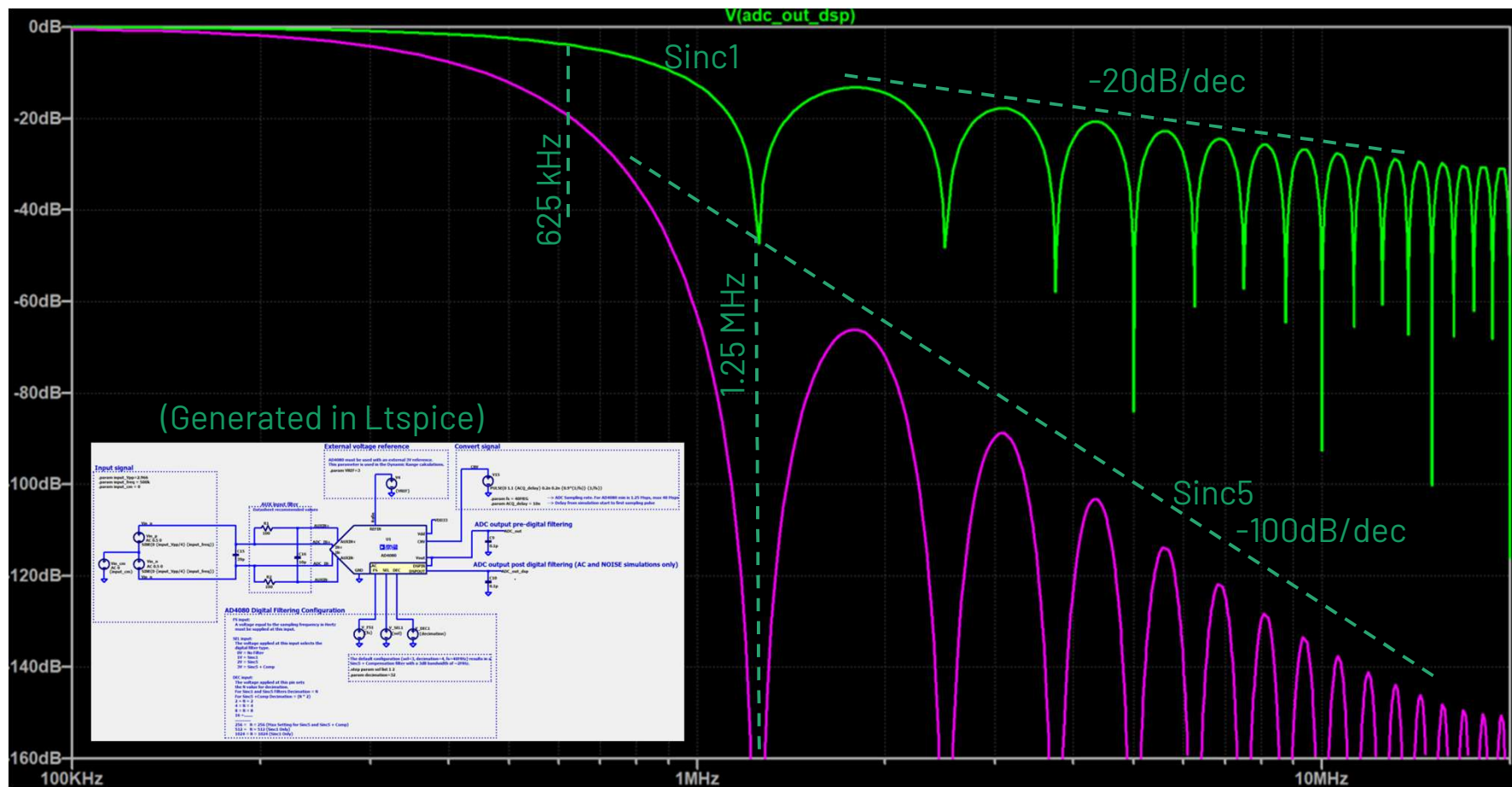
First notch @ f_{sample} / R

1.25 MHz

625 kHz only down ~4 dB ☹



Digital Filter Characteristics – Sinc1 vs. Sinc5 (DF 32)



Disable Digital Filter



Select 'filter_sel'

Select 'none'

The screenshot shows the SCOPE application interface with three numbered callouts indicating the steps to disable the digital filter:

- 1**: The 'Debugger' option in the left sidebar is highlighted with a yellow box.
- 2**: In the central tree view, the 'filter_sel' attribute is highlighted with a yellow box.
- 3**: In the right-hand 'Attributes' panel, the value for 'filter_sel' is set to 'disabled', which is highlighted with a yellow box.

The 'Attributes' panel also displays the following information:

General info

This device is NOT hardware monitor.
This device is NOT trigger.
No trigger on this device.

Name	Value	Type	Path
------	-------	------	------

Stop Signal Generator



Stop Signal Generator



Part 3

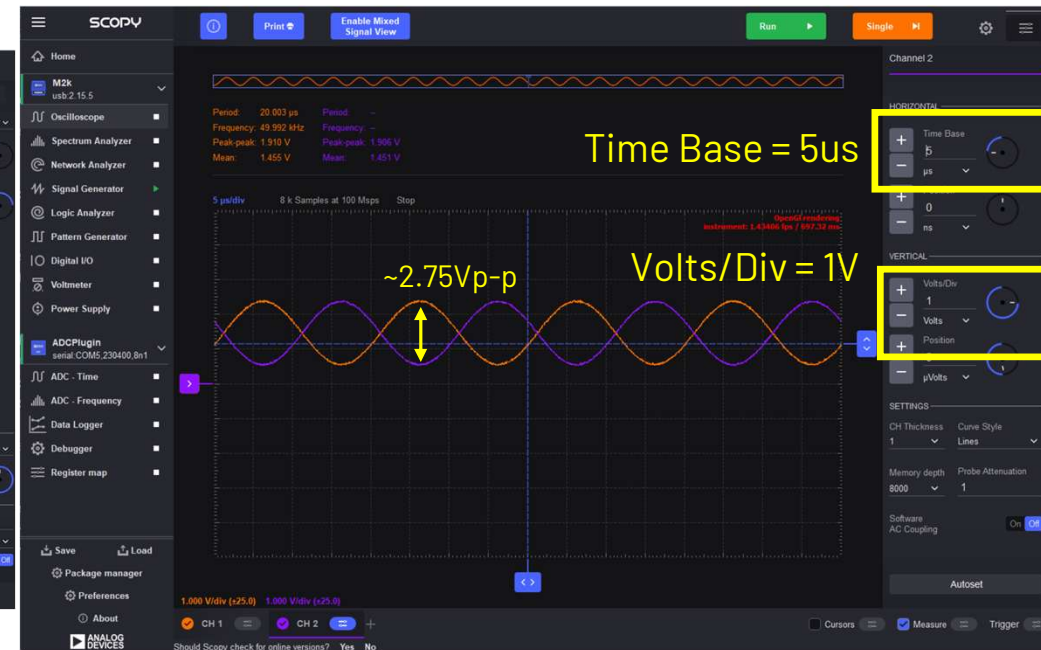
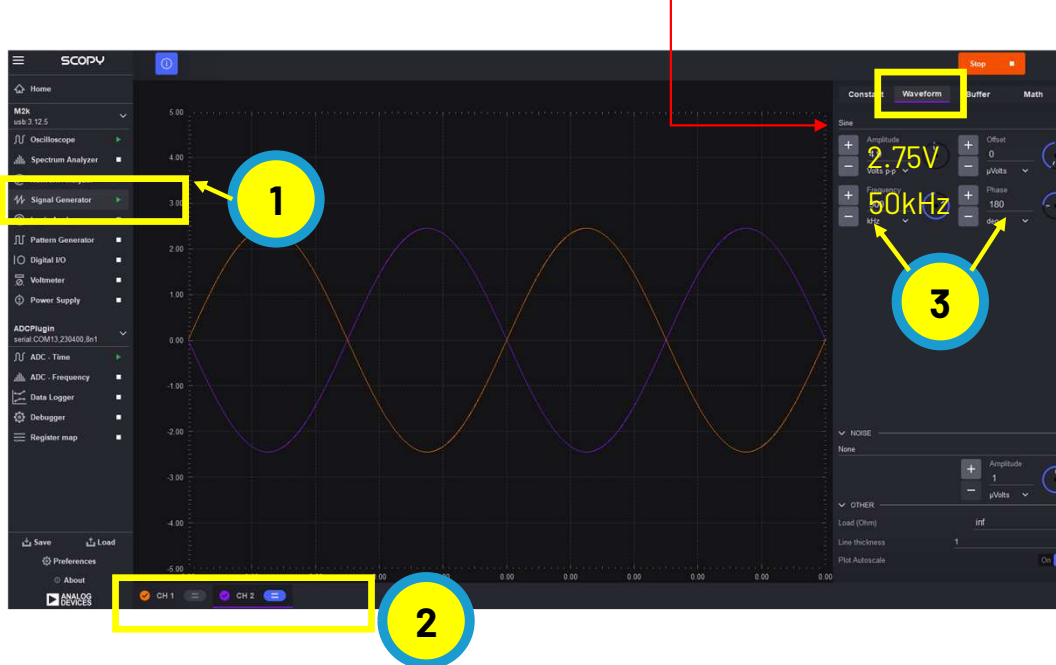
From Sinc1 to Brickwall - seeing
the
Effects in Frequency
Domain

Generate a fully differential signal and measure wrt. GND



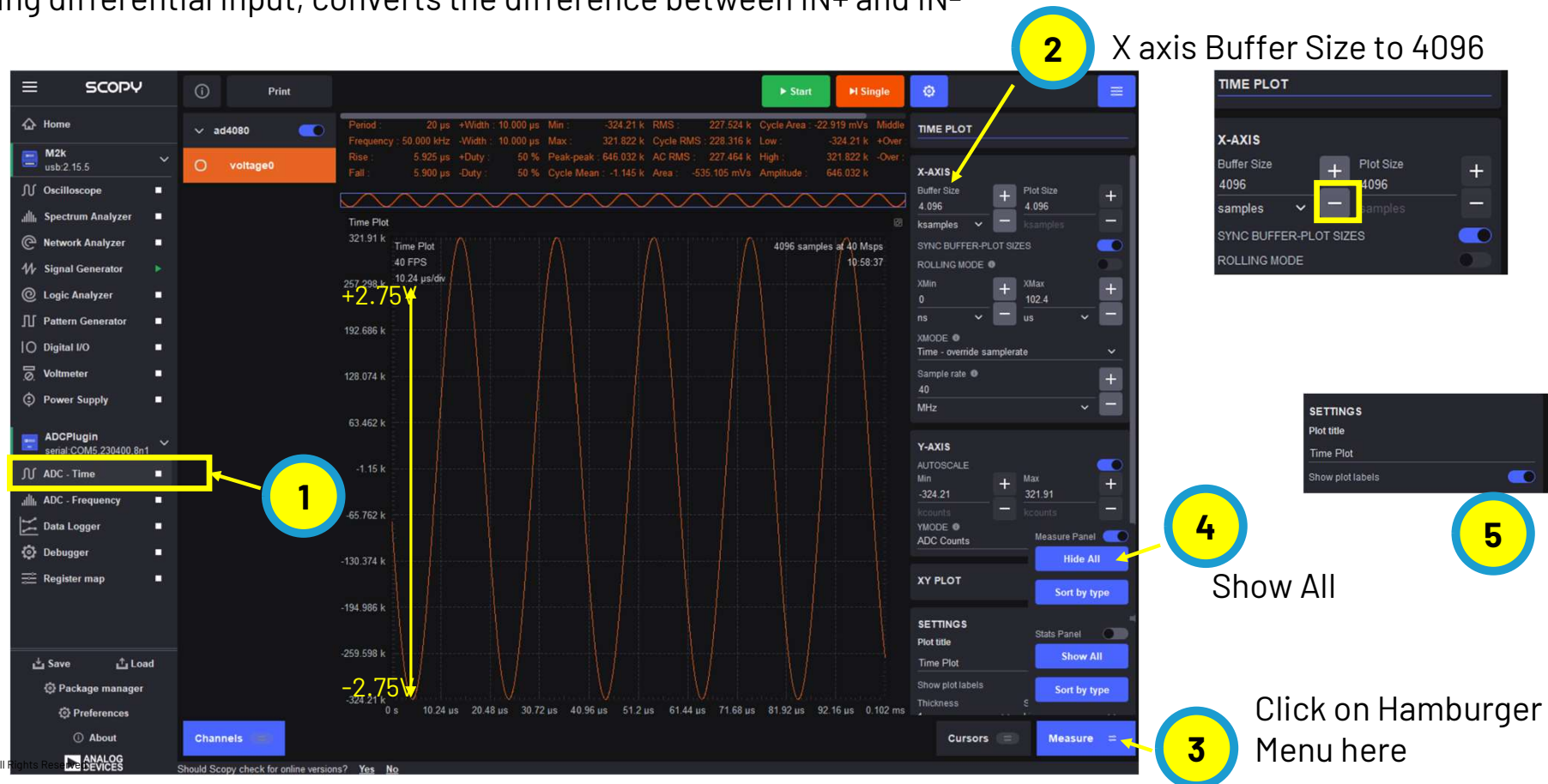
Channel 1 : 2.75 Vpp, 50 kHz Sinewave, Phase = 0 deg
Channel 2 : 2.75 Vpp, 50 kHz Sinewave, Phase = 180 deg

Oscilloscope:
Time Base: 5us
Volts/Div = 1V



ADC - Time

The ADC, being differential input, converts the difference between IN+ and IN-



ADC - Frequency



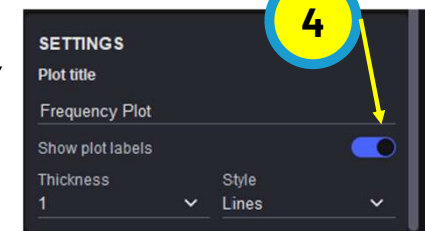
Change Buffer size to:
Buffer size = **16384**

Change Xmode to:
**Frequency - override
samplerate**

Tell GUI of your sample rate
And Nyquist Rate

Xmax = **20MHz**
Sample Rate = **40MHz**

Scroll Down
Enable: Show Plot Labels



Print Me!

Digital Filter impact on the Noise Floor – Sinc1

Effective Sampling Frequency (ODR) = $F_s / \text{dec_rate}$

$F_s \text{ eff (ODR sample rate)} = 40\text{M} / 8 = 5\text{MHz}$

Nyquist Rate (X_{max}) = 2.5MHz



filter_sel = sinc1
sinc_dec_rate = 8

Play with this configuration
to see the frequency
response changing



Sample Rate = 40MHz /
oversampling ratio

Handy guide to output data rate vs. DF ($F_s=40$ Msps)

OSR/DF	ODR (SINC1, SINC5)	ODR (SINC5+PF)
2	20 Msps	10 Msps
4	10 Msps	5 Msps
8	5 Msps	2.5 Msps
16	2.5 Msps	1.25 Msps
32	1.25 Msps	625 ksps
64	625 ksps	312.5 ksps
128	312.5 ksps	156.25
256	156.25	78.125
512	78.125	
1024	39.0625	

Set XMODE to:

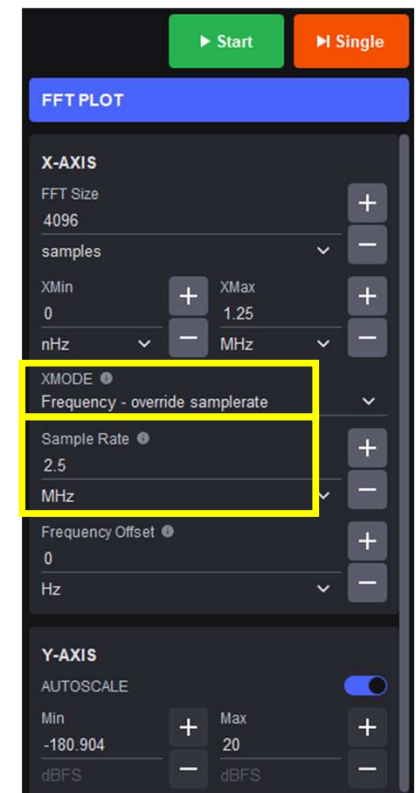
Frequency - override samplerate

Enter ODR in ADC - Frequency instrument's Sample Rate field

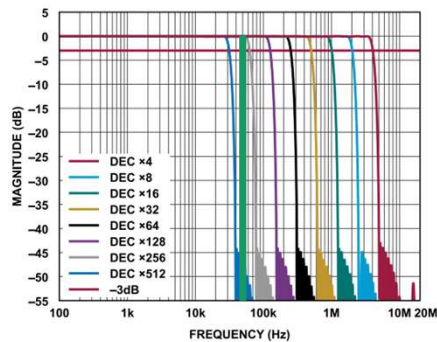
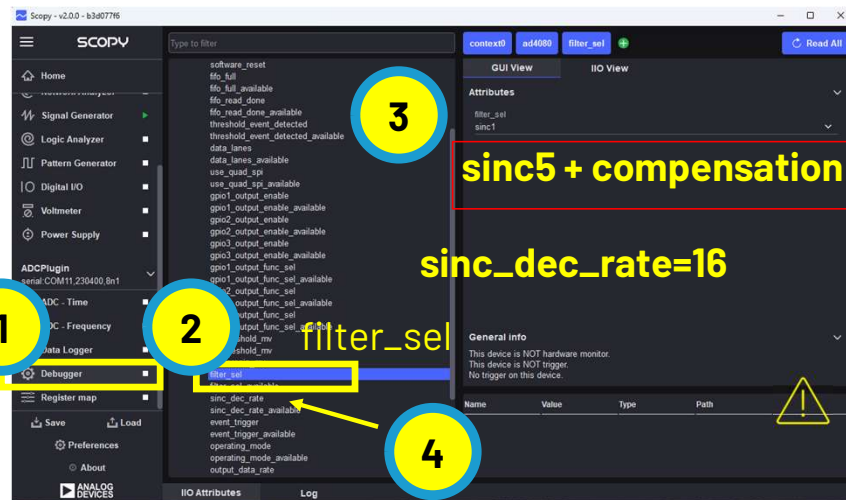
Beware kHz/Mhz units - easiest to type in units shown, hit <<enter>>

Observe: OSR vs. noise floor: 3 dB improvement for every doubling of OSR

Print Me!



sinc5 + compensation for finer adjustment of signal BW



$$\text{ODR}(\text{sinc5} + \text{pf1}) = \frac{f_s}{2 \times \text{oversampling ratio}} = 1.25 \text{ MHz}$$

$$\text{XMAX} = 650 \text{ kHz}$$

Figure 11. Sinc5 + Compensation Filter Response, $f_s = 40 \text{ MHz}$

Extra Credit: Observe SINC5+Comp. step response



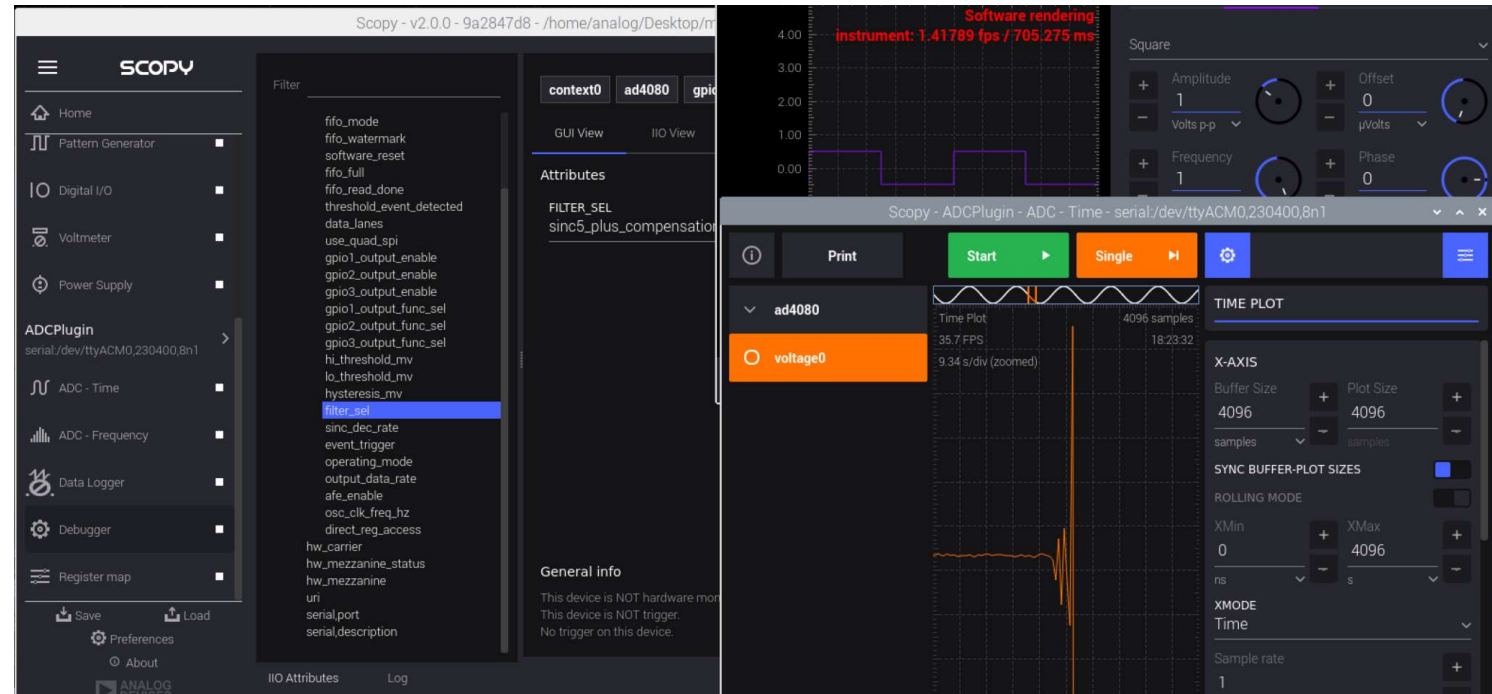
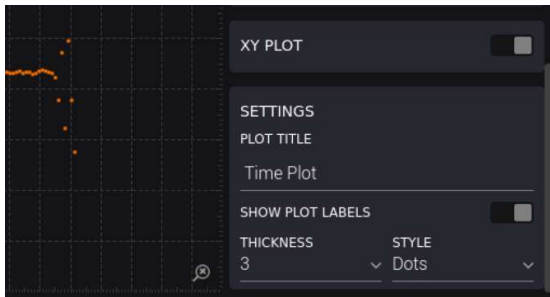
Set FILTER_SEL to:

sinc5_plus_compensation

Set SINC_DEC_RATE to: **16**

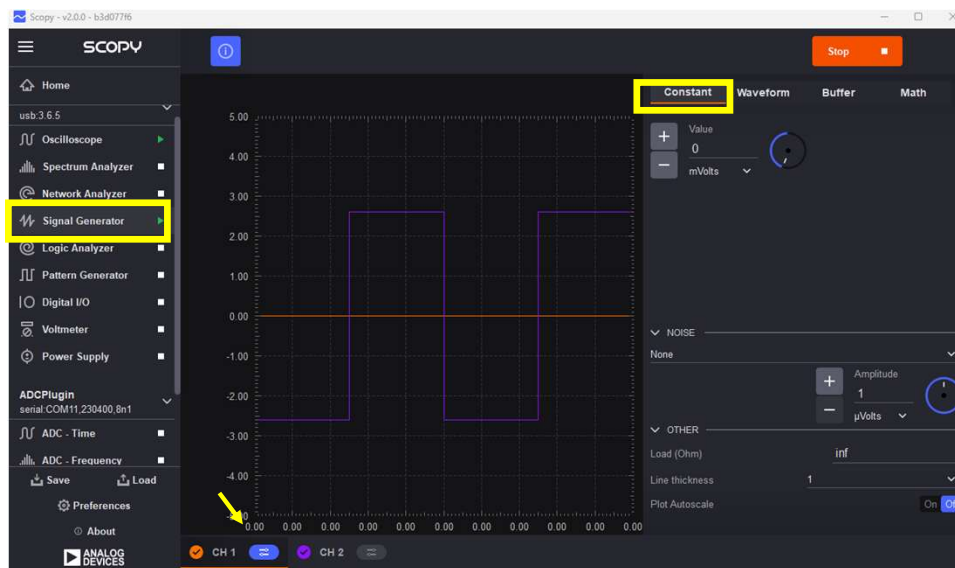
Note that you can zoom in on a region of the waveform by selecting w/ mouse. Right-click to go back.

Also dot display option:

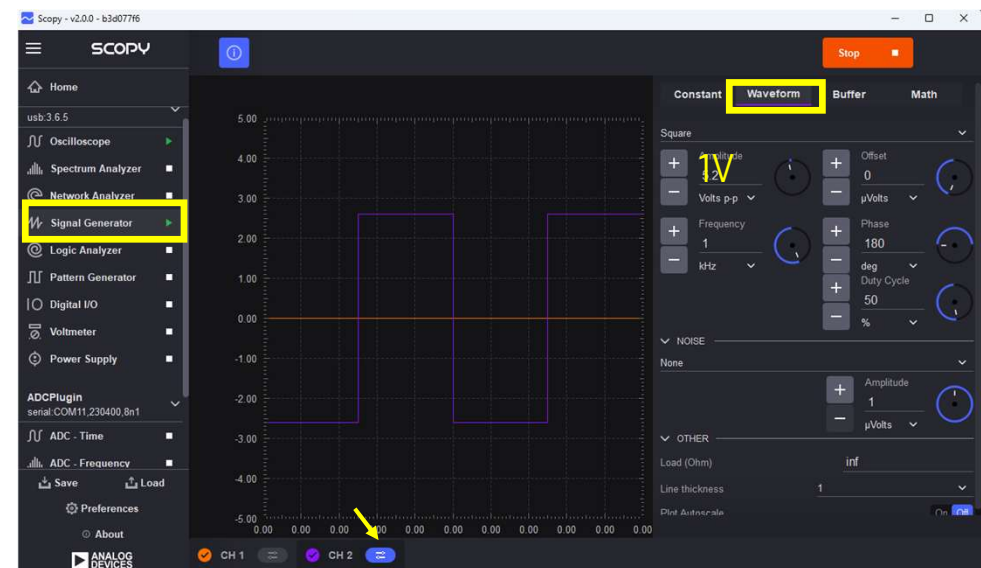


Set Signal Generator

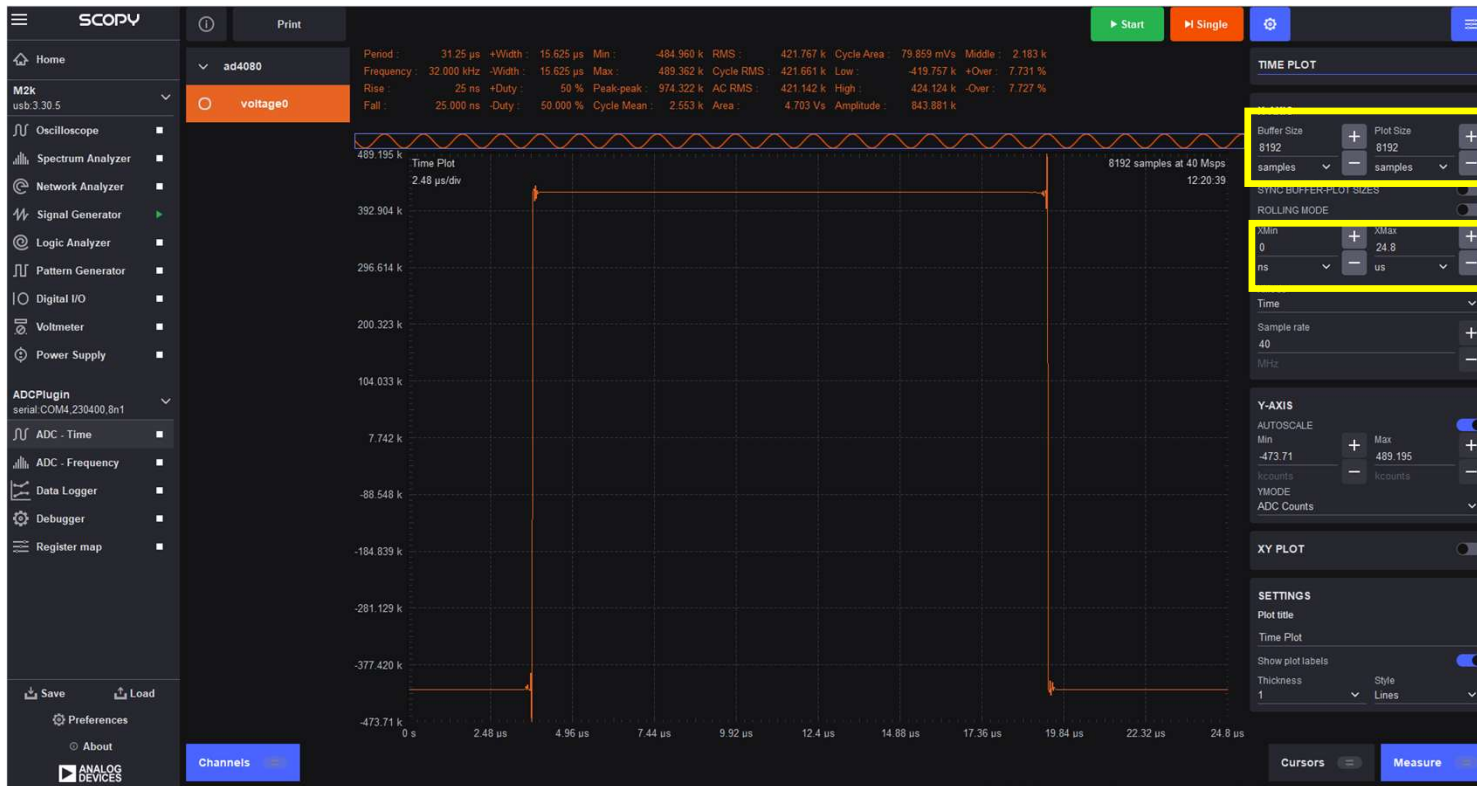
Channel 1.- constant 0V



Channel 2.- 1 kHz square wave
Vpp=1V



Observe the time domain response to the step



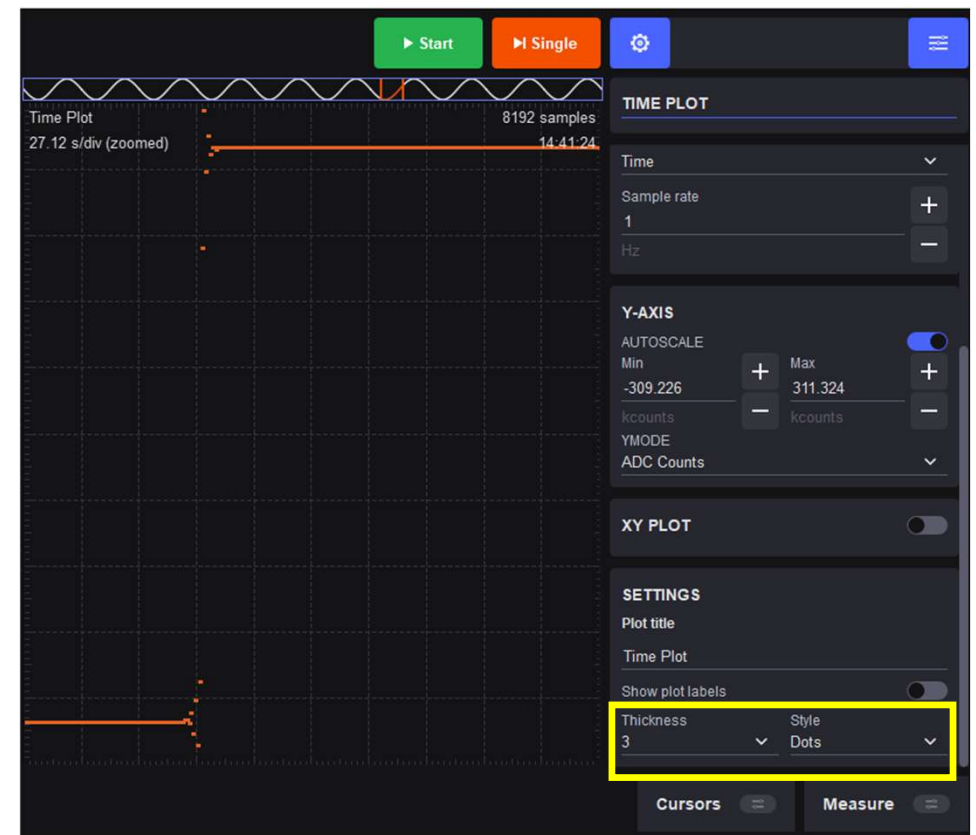
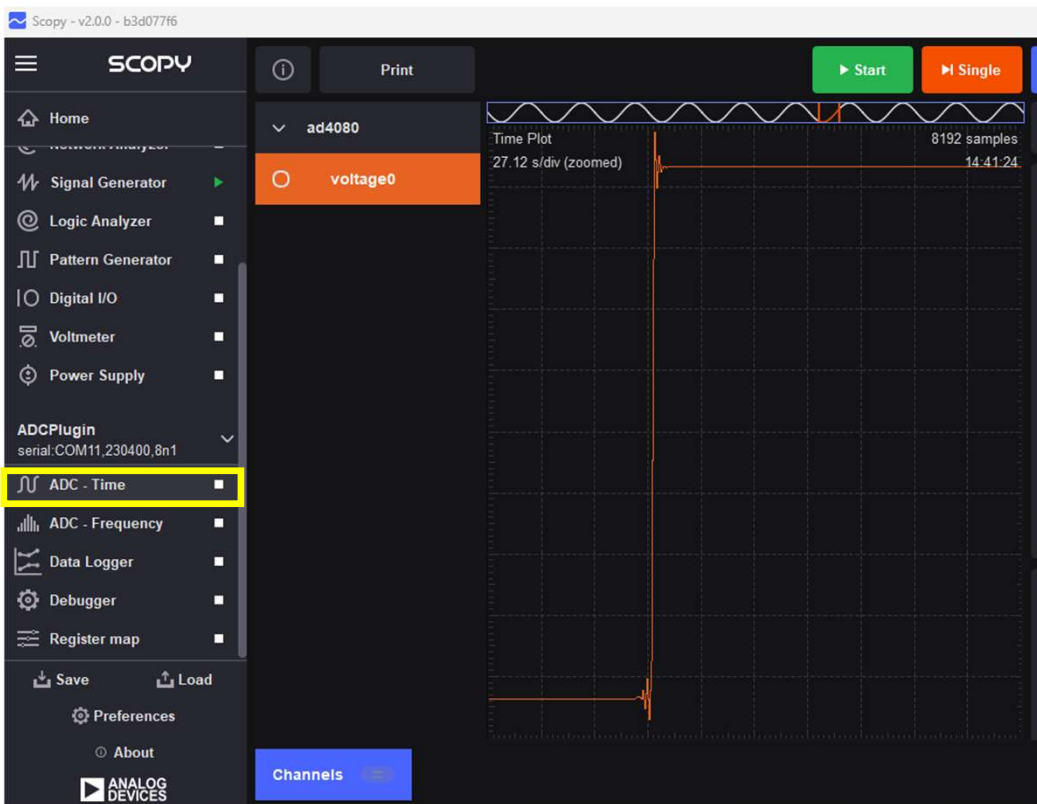
X Axis Buffer Size = 8192

Xmax = 24.8

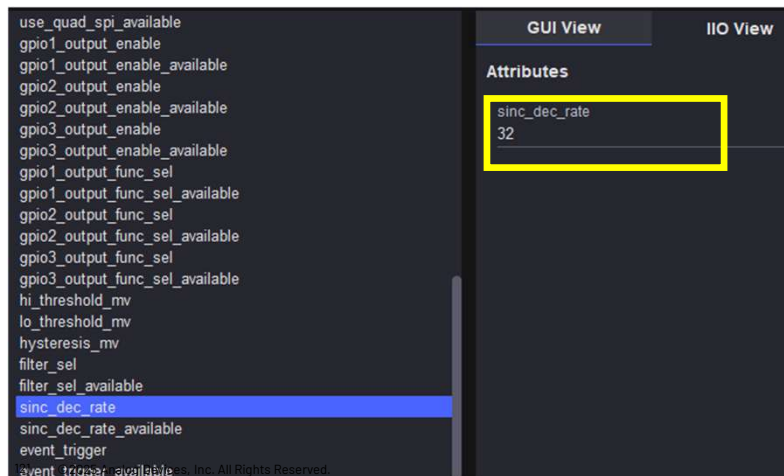
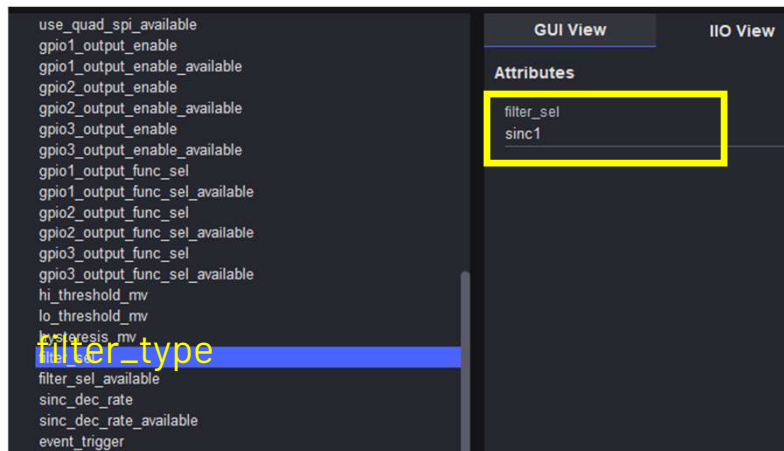
Then Click on Stop to Freeze the capture

Observe the time domain response to the step

Settings -> Style=dots to see actual ADC samples



SINC1 has faster settling time...



AHEAD OF WHAT'S POSSIBLE

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